Eureka Math（A Story of Units）K–5 大纲与海报计划

版本：Eureka Math™ A Story of Units（Great Minds 编写，即纽约州 EngageNY 免费发布的 K–5 数学模块），依据 Great Minds 官网免费提供的 Eureka Basic Curriculum Files 中的各模块 Teacher Edition PDF（版本号 1.3.0–1.3.2，2015–2016 年印次，© 2015 Great Minds，非商业使用依 CC BY-NC-SA 4.0 授权）。模块、主题（Topic）的英文标题逐字摘自各 Teacher Edition 的目录，每节课的 Objective 逐字摘自课时页面，没有自行编写。EngageNY 网站已于 2022 年停止服务，下面也给出了互联网档案馆里的原模块页面。

海报粒度：每个 Topic 一张。全套 214 个 Topic（幼儿园 34、一年级 35、二年级 37、三年级 35、四年级 37、五年级 36），不超过约 250 张，所以按 Topic 出海报。

本地化：英制单位改为公制，美元改为人民币，涉及的 Topic 下面有标注。海报里自己举的例子都标了「示例」。

来源

- Great Minds：A Story of Units: Eureka Basic Curriculum Files（各模块 Teacher Edition PDF，下面每个模块都给了链接）
- 课程总览：A Story of Units Curriculum Overview（PDF）
- EngageNY 原模块页面（互联网档案馆），例如 Grade 3 Module 1

Eureka Math² 的不同之处

Eureka Math²（Great Minds 2021 年起推出的新版）不是同一套教材的小修订，而是重新编排：每个年级统一为 6 个模块（一年级第 6 模块分成两部分），模块里的 Topic 和课时也重新划分了。本套海报按原版 Eureka Math / EngageNY 制作。各年级 Eureka Math² 的模块（摘自 Great Minds 发布的 Eureka Math² 纽约 Next Generation 版 Scope and Sequence 图）：

- **幼儿园**（原版 6 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Counting and Cardinality；M2 Two- and Three-Dimensional Shapes；M3 Comparison；M4 Composition and Decomposition；M5 Addition and Subtraction；M6 Place Value Foundations
- **一年级**（原版 6 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Counting, Comparison, and Addition；M2 Addition and Subtraction Relationships；M3 Properties of Operations to Make Easier Problems；M4 Comparison and Composition of Length Measurements；M5 Place Value Concepts to Compare, Add, and Subtract；M6 Part 1 Attributes of Shapes / Part 2 Advancing Place Value, Addition, and Subtraction
- **二年级**（原版 8 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Place Value Concepts Through Metric Measurement and Data • Place Value, Counting, and Comparing Within 1,000；M2 Addition and Subtraction Within 200；M3 Shapes and Time with Fraction Concepts；M4 Addition and Subtraction Within 1,000；M5 Money, Data, and Customary Measurement；M6 Multiplication and Division Foundations
- **三年级**（原版 7 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Multiplication and Division with Units of 2, 3, 4, 5, and 10；M2 Place Value Concepts Through Metric Measurement；M3 Multiplication and Division with Units of 0, 1, 6, 7, 8, and 9；M4 Multiplication and Area；M5 Fractions as Numbers；M6 Geometry, Measurement, and Data
- **四年级**（原版 7 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Place Value Concepts for Addition and Subtraction；M2 Place Value Concepts for Multiplication and Division；M3 Multiplication and Division of Multi-Digit Numbers；M4 Foundations for Fraction Operations；M5 Angle Measurements and Plane Figures；M6 Place Value Concepts for Decimal Fractions
- **五年级**（原版 6 个模块 → Eureka Math² 6 个模块，Scope and Sequence）：M1 Place Value Concepts for Multiplication and Division with Whole Numbers；M2 Addition and Subtraction with Fractions；M3 Multiplication and Division with Fractions；M4 Place Value Concepts for Decimal Operations；M5 Addition and Multiplication with Area and Volume；M6 Foundations to Geometry in the Coordinate Plane

主要变化举例：幼儿园把原第 5 模块的十几的数移到新第 6 模块“位值基础”；二年级把原第 2 模块（长度）和第 3 模块（1000 以内的数）合并为新第 1 模块，原第 7 模块（长度、钱和数据）在新版里变成第 5 模块“钱、数据和英制测量”；三年级把原第 6 模块（数据）和第 7 模块（几何与测量）合并为新第 6 模块；四年级把原第 2 模块（公制换算）的内容放进新第 1 模块（Topic E 公制测量与换算表），分数从原第 5 模块提前到新第 4 模块；五年级新第 1 模块只讲整数，小数运算集中到新第 4 模块。

幼儿园（Kindergarten）· 6 个模块 · 34 个 Topic

第1模块 10 以内的数（Module 1: Numbers to 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 两个物体的特征（Topic A: Attributes of Two Related Objects）

- Lesson 1：Analyze to find two objects that are exactly the same or not exactly the same.
- Lesson 2：Analyze to find two similar objects—these are the same but…
- Lesson 3：Classify to find two objects that share a visual pattern, color, and use.
- 海报要点：完全一样、不完全一样；按颜色、图案、用途找相同

B主题 分类并计数（Topic B: Classify to Make Categories and Count）

- Lesson 4：Classify items into two pre-determined categories.
- Lesson 5：Classify items into three categories, determine the count in each, and reason about how the last number named determines the total.
- Lesson 6：Sort categories by count. Identify categories with 2, 3, and 4 within a given scenario.
- 海报要点：分成两类、三类，数一数每类有几个

C主题 5 以内的数（Topic C: Numbers to 5 in Different Configurations, Math Drawings, and Expressions）

- Lesson 7：Sort by count in vertical columns and horizontal rows (linear configurations to 5). Match to numerals on cards.
- Lesson 8：Answer how many questions to 5 in linear configurations (5-group), with 4 in an array configuration. Compare ways to count five fingers.
- Lesson 9：Within linear and array dot configurations of numbers 3, 4, and 5, find hidden partners.
- Lesson 10：Within circular and scattered dot configurations of numbers 3, 4, and 5, find hidden partners.
- Lesson 11：Model decompositions of 3 with materials, drawings, and expressions. Represent the decomposition as 1 + 2 and 2 + 1.
- 海报要点：排成一行、点阵、手指；藏起来的部分

D主题 0 的认识与 0–5（Topic D: The Concept of Zero and Working with Numbers 0–5）

- Lesson 12：Understand the meaning of zero. Write the numeral 0.
- Lesson 13：Order and write numerals 0–3 to answer how many questions.
- Lesson 14：Write numerals 1–3. Represent decompositions with materials, drawings, and equations, 3 = 2 + 1 and 3 = 1 + 2.
- Lesson 15：Order and write numerals 4 and 5 to answer how many questions in categories; sort by count.
- Lesson 16：Write numerals 1–5 in order. Answer and make drawings of decompositions with totals of 4 and 5 without equations.
- 海报要点：0 表示没有；写 0–5；3 = 2 + 1

E主题 6–8 的数（Topic E: Working with Numbers 6–8 in Different Configurations）

- Lesson 17：Count 4–6 objects in vertical and horizontal linear configurations and array configurations. Match 6 objects to the numeral 6.
- Lesson 18：Count 4–6 objects in circular and scattered configurations. Count 6 items out of a larger set. Write numerals 1–6 in order.
- Lesson 19：Count 5—7 linking cubes in linear configurations. Match with numeral 7. Count on fingers from 1 to 7, and connect to 5-group images.
- Lesson 20：Reason about sets of 7 varied objects in circular and scattered configurations. Find a path through the scattered configuration. Write numeral 7. Ask, “How is your seven different from mine?”
- Lesson 21：Compare counts of 8 in linear and array configurations. Match with numeral 8.
- Lesson 22：Arrange and strategize to count 8 beans in circular (around a cup) and scattered configurations. Write numeral 8. Find a path through the scatter set, and compare paths with a partner.
- 海报要点：直线、阵列、圆圈、散点排列中数 6、7、8

F主题 9 和 10 的数（Topic F: Working with Numbers 9–10 in Different Configurations）

- Lesson 23：Organize and count 9 varied geometric objects in linear and array (3 threes) configurations. Place objects on 5-group mat. Match with numeral 9.
- Lesson 24：Strategize to count 9 objects in circular (around a paper plate) and scattered configurations printed on paper. Write numeral 9. Represent a path through the scatter count with a pencil. Number each object.
- Lesson 25：Count 10 objects in linear and array configurations (2 fives). Match with numeral 10. Place on the 5-group mat. Dialogue about 9 and 10. Write numeral 10.
- Lesson 26：Count 10 objects in linear and array configurations (2 fives). Match with numeral 10. Place on the 5-group mat. Dialogue about 9 and 10. Write numeral 10.
- Lesson 27：Count 10 objects, and move between all configurations.
- Lesson 28：Act out result unknown story problems without equations.
- 海报要点：两个 5 是 10；在五格板上摆 9 和 10

G主题 多 1（Topic G: One More with Numbers 0–10）

- Lesson 29：Order and match numeral and dot cards from 1 to 10. State 1 more than a given number.
- Lesson 30：Make math stairs from 1 to 10 in cooperative groups.
- Lesson 31：Arrange, analyze, and draw 1 more up to 10 in configurations other than towers.
- Lesson 32：Arrange, analyze, and draw sequences of quantities of 1 more, beginning with numbers other than 1.
- 海报要点：1 到 10 的数学楼梯，每次多 1

H主题 少 1（Topic H: One Less with Numbers 0–10）

- Lesson 33：Order quantities from 10 to 1, and match numerals.
- Lesson 34：Count down from 10 to 1, and state 1 less than a given number.
- Lesson 35：Arrange number towers in order from 10 to 1, and describe the pattern.
- Lesson 36：Arrange, analyze, and draw sequences of quantities that are 1 less in configurations other than towers.

- Lesson 37：Culminating Task Decide how to classify the objects in your bag into two groups. Count the number of objects in each group. Represent the greater number in various ways. Next, remove the card from your pack that shows the number of objects in the smaller group. Put your remaining cards in order from
- 海报要点：从 10 倒数到 1，每次少 1

第2模块 平面图形和立体图形（Module 2: Two-Dimensional and Three-Dimensional Shapes）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 平面图形（Topic A: Two-Dimensional Flat Shapes）

- Lesson 1：Find and describe flat triangles, squares, rectangles, hexagons, and circles using informal language without naming.
- Lesson 2：Explain decisions about classifications of triangles into categories using variants and non-examples. Identify shapes as triangles.
- Lesson 3：Explain decisions about classifications of rectangles into categories using variants and non-examples. Identify shapes as rectangles.
- Lesson 4：Explain decisions about classifications of hexagons and circles, and identify them by name. Make observations using variants and non-examples.
- Lesson 5：Describe and communicate positions of all flat shapes using the words above, below, beside, in front of, next to, and behind.
- 海报要点：三角形、长方形、正方形、六边形、圆；上下前后的位置

B主题 立体图形（Topic B: Three-Dimensional Solid Shapes）

- Lesson 6：Find and describe solid shapes using informal language without naming.
- Lesson 7：Explain decisions about classification of solid shapes into categories. Name the solid shapes.
- Lesson 8：Describe and communicate positions of all solid shapes using the words above, below, beside, in front of, next to, and behind.
- 海报要点：球、正方体、圆柱、圆锥；描述位置

C主题 平面与立体（Topic C: Two-Dimensional and Three-Dimensional Shapes）

- Lesson 9：Identify and sort shapes as two-dimensional or three-dimensional, and recognize two-dimensional and three-dimensional shapes in different orientations and sizes.
- Lesson 10：Culminating task—collaborative groups create displays of different flat shapes with examples, non-examples, and a corresponding solid shape.
- 海报要点：区分平面图形和立体图形；立体图形的面

第3模块 长度、重量、容量和 10 以内的数的比较（Module 3: Comparison of Length, Weight, Capacity, and Numbers to 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 比较长短和高矮（Topic A: Comparison of Length and Height）

- Lesson 1：Compare lengths using taller than and shorter than with aligned and non-aligned endpoints.
- Lesson 2：Compare length measurements with string.

- Lesson 3：Make a series of longer than and shorter than comparisons.
- 海报要点：两端对齐比长短；用绳子比

B主题 用连接方块比长短（Topic B: Comparison of Length and Height of Linking Cube Sticks Within 10）

- Lesson 4：Compare the length of linking cube sticks to a 5-stick.
- Lesson 5：Determine which linking cube stick is longer than or shorter than the other.
- Lesson 6：Compare the length of linking cube sticks to various objects.
- Lesson 7：Compare objects using the same as.
- 海报要点：和 5 个方块的棒比；一样长

C主题 比较轻重（Topic C: Comparison of Weight）

- Lesson 8：Compare using heavier than and lighter than with classroom objects.
- Lesson 9：Compare objects using heavier than, lighter than, and the same as with balance scales.
- Lesson 10：Compare the weight of an object to a set of unit weights on a balance scale.
- Lesson 11：Observe conservation of weight on the balance scale.
- Lesson 12：Compare the weight of an object with sets of different objects on a balance scale.
- 海报要点：天平：重的一边往下沉

D主题 比较容量（Topic D: Comparison of Volume）

- Lesson 13：Compare volume using more than, less than, and the same as by pouring.
- Lesson 14：Explore conservation of volume by pouring.
- Lesson 15：Compare using the same as with units.
- 海报要点：倒一倒：哪个装得多

E主题 够不够（Topic E: Are There Enough?）

- Lesson 16：Make informal comparison of area.
- Lesson 17：Compare to find if there are enough.
- Lesson 18：Compare using more than and the same as.
- Lesson 19：Compare using fewer than and the same as.
- 海报要点：一一对应，看够不够

F主题 比较 10 以内的一组物体（Topic F: Comparison of Sets Within 10）

- Lesson 20：Relate more and less to length.
- Lesson 21：Compare sets informally using more, less, and fewer.
- Lesson 22：Identify and create a set that has the same number of objects.
- Lesson 23：Reason to identify and make a set that has 1 more.
- Lesson 24：Reason to identify and make a set that has 1 less.
- 海报要点：多、少、同样多；多 1、少 1

G主题 比较数字（Topic G: Comparison of Numerals）

- Lesson 25：Match and count to compare a number of objects. State which quantity is more.
- Lesson 26：Match and count to compare two sets of objects. State which quantity is less.
- Lesson 27：Strategize to compare two sets.
- Lesson 28：Visualize quantities to compare two numerals.
- 海报要点：比较两个数，哪个大

H主题 可以测量的特征（Topic H: Clarification of Measurable Attributes）

- Lesson 29：Observe cups of colored water of equal volume poured into a variety of container shapes.
- Lesson 30：Use balls of clay of equal weights to make sculptures.
- Lesson 31：Use benchmarks to create and compare rectangles of different lengths to make a city.
- Lesson 32：Culminating task—describe measurable attributes of single objects.
- 海报要点：同一个物体可以比长短、轻重、容量

第4模块 数对、10 以内的加减法（Module 4: Number Pairs, Addition and Subtraction to 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 2–5 的组成与分解（Topic A: Compositions and Decompositions of 2, 3, 4, and 5）

- Lesson 1：Model composition and decomposition of numbers to 5 using actions, objects, and drawings.
- Lesson 2：Model composition and decomposition of numbers to 5 using fingers and linking cube sticks.
- Lesson 3：Represent composition story situations with drawings using numeric number bonds.
- Lesson 4：Represent decomposition story situations with drawings using numeric number bonds.
- Lesson 5：Represent composition and decomposition of numbers to 5 using pictorial and numeric number bonds.
- Lesson 6：Represent number bonds with composition and decomposition story situations.
- 海报要点：数的分合：数字分合图

B主题 6、7、8 的分解（Topic B: Decompositions of 6, 7, and 8 into Number Pairs）

- Lesson 7：Model decompositions of 6 using a story situation, objects, and number bonds.
- Lesson 8：Model decompositions of 7 using a story situation, sets, and number bonds.
- Lesson 9：Model decompositions of 8 using a story situation, arrays, and number bonds.
- Lesson 10：Model decompositions of 6–8 using linking cube sticks to see patterns.
- Lesson 11：Represent decompositions for 6–8 using horizontal and vertical number bonds.
- Lesson 12：Use 5-groups to represent the 5 + n pattern to 8.
- 海报要点：6、7、8 的所有分法

C主题 和是 6、7、8 的加法（Topic C: Addition with Totals of 6, 7, and 8）

- Lesson 13：Represent decomposition and composition addition stories to 6 with drawings and equations with no unknown.
- Lesson 14：Represent decomposition and composition addition stories to 7 with drawings and equations with no unknown.
- Lesson 15：Represent decomposition and composition addition stories to 8 with drawings and equations with no unknown.
- Lesson 16：Solve add to with result unknown word problems to 8 with equations. Box the unknown.
- Lesson 17：Solve put together with total unknown word problems to 8 using objects and drawings.
- Lesson 18：Solve both addends unknown word problems to 8 to find addition patterns in number pairs.
- 海报要点：加法故事与算式

D主题 8 以内的减法（Topic D: Subtraction from Numbers to 8）

- Lesson 19：Use objects and drawings to find how many are left.
- Lesson 20：Solve take from with result unknown expressions and equations using the minus sign with no unknown.
- Lesson 21：Represent subtraction story problems using objects, drawings, expressions, and equations.
- Lesson 22：Decompose the number 6 using 5-group drawings by breaking off or removing a part, and record each decomposition with a drawing and subtraction equation.
- Lesson 23：Decompose the number 7 using 5-group drawings by hiding a part, and record each decomposition with a drawing and subtraction equation.
- Lesson 24：Decompose the number 8 using 5-group drawings and crossing off a part, and record each decomposition with a drawing and subtraction equation.
- 海报要点：还剩几个：减号

E主题 9 和 10 的分解（Topic E: Decompositions of 9 and 10 into Number Pairs）

- Lesson 25：Model decompositions of 9 using a story situation, objects, and number bonds.
- Lesson 26：Model decompositions of 9 using fingers, linking cubes, and number bonds.
- Lesson 27：Model decompositions of 10 using a story situation, objects, and number bonds.
- Lesson 28：Model decompositions of 10 using fingers, sets, linking cubes, and number bonds.
- 海报要点：9、10 的分合

F主题 和是 9、10 的加法（Topic F: Addition with Totals of 9 and 10）

- Lesson 29：Represent pictorial decomposition and composition addition stories to 9 with 5-group drawings and equations with no unknown.
- Lesson 30：Represent pictorial decomposition and composition addition stories to 10 with 5-group drawings and equations with no unknown.
- Lesson 31：Solve add to with total unknown and put together with total unknown problems with totals of 9 and 10.
- Lesson 32：Solve both addends unknown word problems with totals of 9 and 10 using 5-group drawings.
- 海报要点：五格图与加法算式

G主题 从 9、10 里减（Topic G: Subtraction from 9 and 10）

- Lesson 33：Solve take from equations with no unknown using numbers to 10.
- Lesson 34：Represent subtraction story problems by breaking off, crossing out, and hiding a part.
- Lesson 35：Decompose the number 9 using 5-group drawings, and record each decomposition with a subtraction equation.
- Lesson 36：Decompose the number 10 using 5-group drawings, and record each decomposition with a subtraction equation.
- 海报要点：划掉、拿走、藏起来

H主题 加 0、加 1 与凑 10（Topic H: Patterns with Adding 0 and 1 and Making 10）

- Lesson 37：Add or subtract 0 to get the same number and relate to word problems wherein the same quantity that joins a set, separates.

- Lesson 38：Add 1 to numbers 1–9 to see the pattern of the next number using 5-group drawings and equations.
- Lesson 39：Find the number that makes 10 for numbers 1–9, and record each with a 5-group drawing.
- Lesson 40：Find the number that makes 10 for numbers 1–9, and record each with an addition equation.
- 海报要点：加 0 不变；加 1 是下一个数；凑成 10

第5模块 10–20 的数与数到 100（Module 5: Numbers 10–20 and Counting to 100）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 10 个一和几个一（Topic A: Count 10 Ones and Some Ones）

- Lesson 1：Count straws into piles of ten; count the piles as 10 ones.
- Lesson 2：Count 10 objects within counts of 10 to 20 objects, and describe as 10 ones and __ ones.
- Lesson 3：Count and circle 10 objects within images of 10 to 20 objects, and describe as 10 ones and ___ ones.
- Lesson 4：Count straws the Say Ten way to 19; make a pile for each ten.
- Lesson 5：Count straws the Say Ten way to 20; make a pile for each ten.
- 海报要点：数出 10 根捆成一捆

B主题 组成 11–20（Topic B: Compose Numbers 11–20 from 10 Ones and Some Ones; Represent and Write Teen Numbers）

- Lesson 6：Model with objects and represent numbers 10 to 20 with place value or Hide Zero cards.
- Lesson 7：Model and write numbers 10 to 20 as number bonds.
- Lesson 8：Model teen numbers with materials from abstract to concrete.
- Lesson 9：Draw teen numbers from abstract to pictorial.
- 海报要点：10 和几组成十几；写十几

C主题 分解 11–20 并计数（Topic C: Decompose Numbers 11–20, and Count to Answer “How Many?” Questions in Varied Configurations）

- Lesson 10：Build a Rekenrek to 20.
- Lesson 11：Show, count, and write numbers 11 to 20 in tower configurations increasing by 1—a pattern of 1 larger.
- Lesson 12：Represent numbers 20 to 11 in tower configurations decreasing by 1—a pattern of 1 smaller.
- Lesson 13：Show, count, and write to answer how many questions in linear and array configurations.
- Lesson 14：Show, count, and write to answer how many questions with up to 20 objects in circular configurations.
- 海报要点：算盘、塔形排列；十几的分解

D主题 数到 100（Topic D: Extend the Say Ten and Regular Count Sequence to 100）

- Lesson 15：Count up and down by tens to 100 with Say Ten and regular counting.
- Lesson 16：Count within tens by ones.
- Lesson 17：Count across tens when counting by ones through 40.

- Lesson 18：Count across tens by ones to 100 with and without objects.
- Lesson 19：Explore numbers on the Rekenrek. (Optional)
- 海报要点：10 个 10 个数；一个一个数到 100

E主题 十几的组成与分解（Topic E: Represent and Apply Compositions and Decompositions of Teen Numbers）

- Lesson 20：Represent teen number compositions and decompositions as addition sentences.
- Lesson 21：Represent teen number decompositions as 10 ones and some ones, and find a hidden part.
- Lesson 22：Decompose teen numbers as 10 ones and some ones; compare some ones to compare the teen numbers.
- Lesson 23：Reason about and represent situations, decomposing teen numbers into 10 ones and some ones and composing 10 ones and some ones into a teen number.
- Lesson 24：Culminating Task—Represent teen number decompositions in various ways.
- 海报要点：十几 = 10 + 几

第6模块 分析、比较和组合图形（Module 6: Analyzing, Comparing, and Composing Shapes）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 搭和画图形（Topic A: Building and Drawing Flat and Solid Shapes）

- Lesson 1：Describe the systematic construction of flat shapes using ordinal numbers.
- Lesson 2：Build flat shapes with varying side lengths and record with drawings.
- Lesson 3：Compose solids using flat shapes as a foundation.
- Lesson 4：Describe the relative position of shapes using ordinal numbers.
- 海报要点：用小棒搭三角形、正方形；用平面搭立体

B主题 组合和分解图形（Topic B: Composing and Decomposing Shapes）

- Lesson 5：Compose flat shapes using pattern blocks and drawings.
- Lesson 6：Decompose flat shapes into two or more shapes.
- Lesson 7：Compose simple shapes to form a larger shape described by an outline.
- Lesson 8：Culminating task—review selected topics to create a cumulative year-end project.
- 海报要点：用图形拼图形，把图形分开

一年级（Grade 1）· 6 个模块 · 35 个 Topic

第1模块 10 以内的加减法（Module 1: Sums and Differences to 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 看出藏着的数（Topic A: Embedded Numbers and Decompositions）

- Lesson 1：Analyze and describe embedded numbers (to 10) using 5-groups and number bonds.
- Lesson 2：Reason about embedded numbers in varied configurations using number bonds.
- Lesson 3：See and describe numbers of objects using 1 more within 5-group configurations.
- 海报要点：五格图里看出几和几；数的分合

B主题 接着数（Topic B: Counting On from Embedded Numbers）

- Lesson 4：Represent put together situations with number bonds. Count on from one embedded number or part to totals of 6 and 7, and generate all addition expressions for each total.
- Lesson 5：Represent put together situations with number bonds. Count on from one embedded number or part to totals of 6 and 7, and generate all addition expressions for each total.
- Lesson 6：Represent put together situations with number bonds. Count on from one embedded number or part to totals of 8 and 9, and generate all expressions for each total.
- Lesson 7：Represent put together situations with number bonds. Count on from one embedded number or part to totals of 8 and 9, and generate all expressions for each total.
- Lesson 8：Represent all the number pairs of 10 as number bonds from a given scenario, and generate all expressions equal to 10.
- 海报要点：从一部分接着往后数

C主题 加法应用题（Topic C: Addition Word Problems）

- Lesson 9：Solve add to with result unknown and put together with result unknown math stories by drawing, writing equations, and making statements of the solution.
- Lesson 10：Solve put together with result unknown math stories by drawing and using 5-group cards.
- Lesson 11：Solve add to with change unknown math stories as a context for counting on by drawing, writing equations, and making statements of the solution.
- Lesson 12：Solve add to with change unknown math stories using 5-group cards.
- Lesson 13：Tell put together with result unknown, add to with result unknown, and add to with change unknown stories from equations.
- 海报要点：合起来、添上；画图解题

D主题 接着数的方法（Topic D: Strategies for Counting On）

- Lesson 14：Count on up to 3 more using numeral and 5-group cards and fingers to track the change.
- Lesson 15：Count on up to 3 more using numeral and 5-group cards and fingers to track the change.
- Lesson 16：Count on to find the unknown part in missing addend equations such as 6 + __ = 9. Answer, “How many more to make 6, 7, 8, 9, and 10?”
- 海报要点：用手指记录多数几个；6 + _ = 9

E主题 交换律和等号（Topic E: The Commutative Property of Addition and the Equal Sign）

- Lesson 17：Understand the meaning of the equal sign by pairing equivalent expressions and constructing true number sentences.
- Lesson 18：Understand the meaning of the equal sign by pairing equivalent expressions and constructing true number sentences.
- Lesson 19：Represent the same story scenario with addends repositioned (the commutative property).
- Lesson 20：Apply the commutative property to count on from a larger addend.
- 海报要点：等号两边一样多；3 + 5 = 5 + 3

F主题 10 以内加法熟练（Topic F: Development of Addition Fluency Within 10）

- Lesson 21：Visualize and solve doubles and doubles plus 1 with 5-group cards.
- Lesson 22：Look for and make use of repeated reasoning on the addition chart by solving and analyzing problems with common addends.
- Lesson 23：Look for and make use of structure on the addition chart by looking for and coloring problems with the same total.
- Lesson 24：Practice to build fluency with facts to 10.
- 海报要点：双数、双数加 1；加法表

G主题 减法是求未知加数（Topic G: Subtraction as an Unknown Addend Problem）

- Lesson 25：Solve add to with change unknown math stories with addition, and relate to subtraction. Model with materials, and write corresponding number sentences.
- Lesson 26：Count on using the number path to find an unknown part.
- Lesson 27：Count on using the number path to find an unknown part.
- 海报要点：9 − 6 想 6 + _ = 9

H主题 减法应用题（Topic H: Subtraction Word Problems）

- Lesson 28：Solve take from with result unknown math stories with math drawings, true number sentences, and statements, using horizontal marks to cross off what is taken away.
- Lesson 29：Solve take apart with addend unknown math stories with math drawings, equations, and statements, circling the known part to find the unknown.
- Lesson 30：Solve add to with change unknown math stories with drawings, relating addition and subtraction.
- Lesson 31：Solve take from with change unknown math stories with drawings.
- Lesson 32：Solve put together/take apart with addend unknown math stories.
- 海报要点：拿走、分开、比较

I主题 减法的分解策略（Topic I: Decomposition Strategies for Subtraction）

- Lesson 33：Model 0 less and 1 less pictorially and as subtraction number sentences.
- Lesson 34：Model n – n and n – (n – 1) pictorially and as subtraction sentences.
- Lesson 35：Relate subtraction facts involving fives and doubles to corresponding decompositions.
- Lesson 36：Relate subtraction from 10 to corresponding decompositions.
- Lesson 37：Relate subtraction from 9 to corresponding decompositions.
- 海报要点：减 0、减 1、减自己；从 10 减

J主题 10 以内减法熟练（Topic J: Development of Subtraction Fluency Within 10）

- Lesson 38：Look for and make use of repeated reasoning and structure using the addition chart to solve subtraction problems.
- Lesson 39：Analyze the addition chart to create sets of related addition and subtraction facts.
- 海报要点：用加法表找相关的加减法

第2模块 20 以内加减法与位值初步（Module 2: Introduction to Place Value Through Addition and Subtraction Within 20）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 凑十法（Topic A: Counting On or Making Ten to Solve Result Unknown and Total Unknown Problems）

- Lesson 1：Solve word problems with three addends, two of which make ten.
- Lesson 2：Use the associative and commutative properties to make ten with three addends.
- Lesson 3：Make ten when one addend is 9.
- Lesson 4：Make ten when one addend is 9.
- Lesson 5：Compare efficiency of counting on and making ten when one addend is 9.
- Lesson 6：Use the commutative property to make ten.
- Lesson 7：Make ten when one addend is 8.
- Lesson 8：Make ten when one addend is 8.
- Lesson 9：Compare efficiency of counting on and making ten when one addend is 8.
- Lesson 10：Solve problems with addends of 7, 8, and 9.
- Lesson 11：Share and critique peer solution strategies for put together with total unknown word problems.
- 海报要点：9 + 5：把 5 分成 1 和 4，凑成 10

B主题 破十法（Topic B: Counting On or Taking from Ten to Solve Result Unknown and Total Unknown Problems）

- Lesson 12：Solve word problems with subtraction of 9 from 10.
- Lesson 13：Solve word problems with subtraction of 9 from 10.
- Lesson 14：Model subtraction of 9 from teen numbers.
- Lesson 15：Model subtraction of 9 from teen numbers.
- Lesson 16：Relate counting on to making ten and taking from ten.
- Lesson 17：Model subtraction of 8 from teen numbers.
- Lesson 18：Model subtraction of 8 from teen numbers.
- Lesson 19：Compare efficiency of counting on and taking from ten.
- Lesson 20：Subtract 7, 8, and 9 from teen numbers.
- Lesson 21：Share and critique peer solution strategies for take from with result unknown and take apart with addend unknown word problems from the teens.
- 海报要点：14 − 9：从 10 里减 9，再加 4

C主题 求未知部分（Topic C: Strategies for Solving Change or Addend Unknown Problems）

- Lesson 22：Solve put together/take apart with addend unknown word problems, and relate counting on to the take from ten strategy.

- Lesson 23：Solve add to with change unknown problems, relating varied addition and subtraction strategies.
- Lesson 24：Strategize to solve take from with change unknown problems.
- Lesson 25：Strategize and apply understanding of the equal sign to solve equivalent expressions.
- 海报要点：加数未知、变化未知的问题

D主题 十几是 1 个十和几个一（Topic D: Varied Problems with Decompositions of Teen Numbers as 1 Ten and Some Ones）

- Lesson 26：Identify 1 ten as a unit by renaming representations of 10.
- Lesson 27：Solve addition and subtraction problems decomposing and composing teen numbers as 1 ten and some ones.
- Lesson 28：Solve addition problems using ten as a unit, and write two-step solutions.
- Lesson 29：Solve subtraction problems using ten as a unit, and write two-step solutions.
- 海报要点：把 10 看成 1 个十

第3模块 长度的排序和比较（Module 3: Ordering and Comparing Length Measurements as Numbers）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 间接比较长度（Topic A: Indirect Comparison in Length Measurement）

- Lesson 1：Compare length directly and consider the importance of aligning endpoints.
- Lesson 2：Compare length using indirect comparison by finding objects longer than, shorter than, and equal in length to that of a string.
- Lesson 3：Order three lengths using indirect comparison.
- 海报要点：两端对齐；用第三个物体比较

B主题 标准长度单位（Topic B: Standard Length Units）

- Lesson 4：Express the length of an object using centimeter cubes as length units to measure with no gaps or overlaps.
- Lesson 5：Rename and measure with centimeter cubes, using their standard unit name of centimeters.
- Lesson 6：Order, measure, and compare the length of objects before and after measuring with centimeter cubes, solving compare with difference unknown word problems.
- 海报要点：用厘米方块量长度

C主题 非标准与标准单位（Topic C: Non-Standard and Standard Length Units）

- Lesson 7：Measure the same objects from Topic B with different non-standard units simultaneously to see the need to measure with a consistent unit.
- Lesson 8：Understand the need to use the same units when comparing measurements with others.
- Lesson 9：Answer compare with difference unknown problems about lengths of two different objects measured in centimeters.
- 海报要点：用不同单位量同一物体；单位要一样

D主题 数据的解读（Topic D: Data Interpretation）

- Lesson 10：Collect, sort, and organize data; then ask and answer questions about the number of data points.

- Lesson 11：Collect, sort, and organize data; then ask and answer questions about the number of data points.
- Lesson 12：Ask and answer varied word problem types about a data set with three categories.
- Lesson 13：Ask and answer varied word problem types about a data set with three categories.
- 海报要点：整理数据；三类数据回答问题

第4模块 40 以内的数位、比较与加减法（Module 4: Place Value, Comparison, Addition and Subtraction to 40）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 十和一（Topic A: Tens and Ones）

- Lesson 1：Compare the efficiency of counting by ones and counting by tens.
- Lesson 2：Use the place value chart to record and name tens and ones within a two-digit number.
- Lesson 3：Interpret two-digit numbers as either tens and some ones or as all ones.
- Lesson 4：Write and interpret two-digit numbers as addition sentences that combine tens and ones.
- Lesson 5：Identify 10 more, 10 less, 1 more, and 1 less than a two-digit number.
- Lesson 6：Use dimes and pennies as representations of tens and ones.
- 海报要点：数位表；1 个十和 4 个一是 14

B主题 比较两位数（Topic B: Comparison of Pairs of Two-Digit Numbers）

- Lesson 7：Compare two quantities, and identify the greater or lesser of the two given numerals.
- Lesson 8：Compare quantities and numerals from left to right.
- Lesson 9：Use the symbols >, =, and < to compare quantities and numerals.
- Lesson 10：Use the symbols >, =, and < to compare quantities and numerals.
- 海报要点：先比十位；用 >、<、=

C主题 整十的加减（Topic C: Addition and Subtraction of Tens）

- Lesson 11：Add and subtract tens from a multiple of 10.
- Lesson 12：Add tens to a two-digit number.
- 海报要点：30 + 10，40 − 20

D主题 两位数加整十或一位数（Topic D: Addition of Tens or Ones to a Two-Digit Number）

- Lesson 13：Use counting on and the make ten strategy when adding across a ten.
- Lesson 14：Use counting on and the make ten strategy when adding across a ten.
- Lesson 15：Use single-digit sums to support solutions for analogous sums to 40.
- Lesson 16：Add ones and ones or tens and tens.
- Lesson 17：Add ones and ones or tens and tens.
- Lesson 18：Share and critique peer strategies for adding two-digit numbers.
- 海报要点：凑十；十加十、一加一

E主题 20 以内的各类问题（Topic E: Varied Problem Types Within 20）

- Lesson 19：Use tape diagrams as representations to solve put together/take apart with total unknown and add to with result unknown word problems.
- Lesson 20：Recognize and make use of part–whole relationships within tape diagrams when solving a variety of problem types.

- Lesson 21：Recognize and make use of part–whole relationships within tape diagrams when solving a variety of problem types.
- Lesson 22：Write word problems of varied types.
- 海报要点：用条形图分析数量关系

F主题 两位数加两位数（Topic F: Addition of Tens and Ones to a Two-Digit Number）

- Lesson 23：Interpret two-digit numbers as tens and ones, including cases with more than 9 ones.
- Lesson 24：Add a pair of two-digit numbers when the ones digits have a sum less than or equal to 10.
- Lesson 25：Add a pair of two-digit numbers when the ones digits have a sum less than or equal to 10.
- Lesson 26：Add a pair of two-digit numbers when the ones digits have a sum greater than 10.
- Lesson 27：Add a pair of two-digit numbers when the ones digits have a sum greater than 10.
- Lesson 28：Add a pair of two-digit numbers with varied sums in the ones.
- Lesson 29：Add a pair of two-digit numbers with varied sums in the ones.
- 海报要点：十位加十位，个位加个位；满十进一

第5模块 认识、组合和平分图形（Module 5: Identifying, Composing, and Partitioning Shapes）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 图形的特征（Topic A: Attributes of Shapes）

- Lesson 1：Classify shapes based on defining attributes using examples, variants, and non-examples.
- Lesson 2：Find and name two-dimensional shapes including trapezoid, rhombus, and a square as a special rectangle, based on defining attributes of sides and corners.
- Lesson 3：Find and name three-dimensional shapes including cone and rectangular prism, based on defining attributes of faces and points.
- 海报要点：边、角决定图形；梯形、菱形；立体图形

B主题 组合图形（Topic B: Part–Whole Relationships Within Composite Shapes）

- Lesson 4：Create composite shapes from two-dimensional shapes.
- Lesson 5：Compose a new shape from composite shapes.
- Lesson 6：Create a composite shape from three-dimensional shapes and describe the composite shape using shape names and positions.
- 海报要点：用图形拼成新图形

C主题 长方形和圆的二分之一和四分之一（Topic C: Halves and Quarters of Rectangles and Circles）

- Lesson 7：Name and count shapes as parts of a whole, recognizing relative sizes of the parts.
- Lesson 8：Partition shapes and identify halves and quarters of circles and rectangles.
- Lesson 9：Partition shapes and identify halves and quarters of circles and rectangles.
- 海报要点：平均分成 2 份、4 份

D主题 用二分之一认识时间（Topic D: Application of Halves to Tell Time）

- Lesson 10：Construct a paper clock by partitioning a circle and tell time to the hour.
- Lesson 11：Recognize halves within a circular clock face and tell time to the half hour.
- Lesson 12：Recognize halves within a circular clock face and tell time to the half hour.

- Lesson 13：Recognize halves within a circular clock face and tell time to the half hour.
- 海报要点：整时与半时

第6模块 100 以内的数位、比较与加减法（Module 6: Place Value, Comparison, Addition and Subtraction to 100）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 比较应用题（Topic A: Comparison Word Problems）

- Lesson 1：Solve compare with difference unknown problem types.
- Lesson 2：Solve compare with bigger or smaller unknown problem types.
- 海报要点：比多少：差未知、大数未知、小数未知

B主题 120 以内的数（Topic B: Numbers to 120）

- Lesson 3：Use the place value chart to record and name tens and ones within a two-digit number up to 100.
- Lesson 4：Write and interpret two-digit numbers to 100 as addition sentences that combine tens and ones.
- Lesson 5：Identify 10 more, 10 less, 1 more, and 1 less than a two-digit number within 100.
- Lesson 6：Use the symbols >, =, and < to compare quantities and numerals to 100.
- Lesson 7：Count and write numbers to 120. Use Hide Zero cards to relate numbers 0 to 20 to 100 to 120.
- Lesson 8：Count to 120 in unit form using only tens and ones. Represent numbers to 120 as tens and ones on the place value chart.
- Lesson 9：Represent up to 120 objects with a written numeral.
- 海报要点：数位表；多 10、少 10；数到 120

C主题 100 以内的加法（Topic C: Addition to 100 Using Place Value Understanding）

- Lesson 10：Add and subtract multiples of 10 from multiples of 10 to 100, including dimes.
- Lesson 11：Add a multiple of 10 to any two-digit number within 100.
- Lesson 12：Add a pair of two-digit numbers when the ones digits have a sum less than or equal to 10.
- Lesson 13：Add a pair of two-digit numbers when the ones digits have a sum greater than 10 using decomposition.
- Lesson 14：Add a pair of two-digit numbers when the ones digits have a sum greater than 10 using decomposition.
- Lesson 15：Add a pair of two-digit numbers when the ones digits have a sum greater than 10 with drawing. Record the total below.
- Lesson 16：Add a pair of two-digit numbers when the ones digits have a sum greater than 10 with drawing. Record the new ten below.
- Lesson 17：Add a pair of two-digit numbers when the ones digits have a sum greater than 10 with drawing. Record the new ten below.
- 海报要点：加整十；两位数加两位数

D主题 多种加法策略（Topic D: Varied Place Value Strategies for Addition to 100）

- Lesson 18：Add a pair of two-digit numbers with varied sums in the ones, and compare the results of different recording methods.
- Lesson 19：Solve and share strategies for adding two-digit numbers with varied sums.
- 海报要点：分享不同的加法算法

E主题 硬币和面值（Topic E: Coins and Their Values）

- Lesson 20：Identify pennies, nickels, and dimes by their image, name, or value. Decompose the values of nickels and dimes using pennies and nickels.
- Lesson 21：Identify quarters by their image, name, or value. Decompose the value of a quarter using pennies, nickels, and dimes.
- Lesson 22：Identify varied coins by their image, name, or value. Add one cent to the value of any coin.
- Lesson 23：Count on using pennies from any single coin.
- Lesson 24：Use dimes and pennies as representations of numbers to 120.
- 海报要点：人民币：1 角、5 角、1 元；凑钱
- 本地化：原书为美国硬币（1、5、10、25 美分），本海报改为人民币（1 角、5 角、1 元）。

F主题 20 以内的各类问题（Topic F: Varied Problem Types Within 20）

- Lesson 25：Solve compare with bigger or smaller unknown problem types.
- Lesson 26：Solve compare with bigger or smaller unknown problem types.
- Lesson 27：Share and critique peer strategies for solving problems of varied types.
- 海报要点：比较问题的多种类型

G主题 学年总结（Topic G: Culminating Experiences）

- Lesson 28：Celebrate progress in fluency with adding and subtracting within 10 (and 20). Organize engaging summer practice.
- Lesson 29：Celebrate progress in fluency with adding and subtracting within 10 (and 20). Organize engaging summer practice.
- Lesson 30：Create folder covers for work to be taken home illustrating the year’ s learning.
- 海报要点：庆祝一年的进步

二年级（Grade 2）· 8 个模块 · 37 个 Topic

第1模块 100 以内的加减法（Module 1: Sums and Differences to 100）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 加减熟练的基础（Topic A: Foundations for Fluency with Sums and Differences Within 100）

- Lesson 1：Practice making ten and adding to ten.
- Lesson 2：Practice making the next ten and adding to a multiple of ten.
- 海报要点：凑十；凑到下一个整十

B主题 100 以内加减初步（Topic B: Initiating Fluency with Addition and Subtraction Within 100）

- Lesson 3：Add and subtract like units.
- Lesson 4：Make a ten to add within 20.
- Lesson 5：Make a ten to add within 100.
- Lesson 6：Subtract single-digit numbers from multiples of 10 within 100.
- Lesson 7：Take from 10 within 20.
- Lesson 8：Take from 10 within 100.
- 海报要点：同单位加减；凑十、破十

第2模块 长度单位的加减（Module 2: Addition and Subtraction of Length Units）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 尺子的概念（Topic A: Understand Concepts About the Ruler）

- Lesson 1：Connect measurement with physical units by using multiple copies of the same physical unit to measure.
- Lesson 2：Use iteration with one physical unit to measure.
- Lesson 3：Apply concepts to create unit rulers and measure lengths using unit rulers.
- 海报要点：用同样的单位连续摆；自制尺子

B主题 用不同工具量和估（Topic B: Measure and Estimate Length Using Different Measurement Tools）

- Lesson 4：Measure various objects using centimeter rulers and meter sticks.
- Lesson 5：Develop estimation strategies by applying prior knowledge of length and using mental benchmarks.
- 海报要点：厘米尺和米尺；估长度

C主题 用不同单位量和比（Topic C: Measure and Compare Lengths Using Different Length Units）

- Lesson 6：Measure and compare lengths using centimeters and meters.
- Lesson 7：Measure and compare lengths using standard metric length units and non-standard length units; relate measurement to unit size.
- 海报要点：厘米与米；单位越大，数越小

D主题 长度与加减法（Topic D: Relate Addition and Subtraction to Length）

- Lesson 8：Solve addition and subtraction word problems using the ruler as a number line.

- Lesson 9：Measure lengths of string using measurement tools, and use tape diagrams to represent and compare lengths.
- Lesson 10：Apply conceptual understanding of measurement by solving two-step word problems.
- 海报要点：把尺子看成数线；条形图

第3模块 1000 以内数的位值、计数与比较（Module 3: Place Value, Counting, and Comparison of Numbers to 1,000）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 一、十、百、千（Topic A: Forming Base Ten Units of Ten, a Hundred, and a Thousand）

- Lesson 1：Bundle and count ones, tens, and hundreds to 1,000.
- 海报要点：10 个一是 1 个十……10 个百是 1 个千

B主题 个、十、百的位值（Topic B: Understanding Place Value Units of One, Ten, and a Hundred）

- Lesson 2：Count up and down between 100 and 220 using ones and tens.
- Lesson 3：Count up and down between 90 and 1,000 using ones, tens, and hundreds.
- 海报要点：按一、十、百数数

C主题 三位数的几种写法（Topic C: Three-Digit Numbers in Unit, Standard, Expanded, and Word Forms）

- Lesson 4：Count up to 1,000 on the place value chart.
- Lesson 5：Write base ten three-digit numbers in unit form; show the value of each digit.
- Lesson 6：Write base ten numbers in expanded form.
- Lesson 7：Write, read, and relate base ten numbers in all forms.
- 海报要点：单位形式、标准形式、展开式、文字

D主题 用人民币表示三位数（Topic D: Modeling Base Ten Numbers Within 1,000 with Money）

- Lesson 8：Count the total value of $1, $10, and $100 bills up to $1,000.
- Lesson 9：Count from $10 to $1,000 on the place value chart and the empty number line.
- Lesson 10：Explore $1,000. How many $10 bills can we change for a thousand dollar bill?
- 海报要点：1 元、10 元、100 元纸币
- 本地化：原书为 1、10、100 美元纸币，本海报改为 1 元、10 元、100 元人民币。

E主题 用数位圆片表示三位数（Topic E: Modeling Numbers Within 1,000 with Place Value Disks）

- Lesson 11：Count the total value of ones, tens, and hundreds with place value disks.
- Lesson 12：Change 10 ones for 1 ten, 10 tens for 1 hundred, and 10 hundreds for 1 thousand.
- Lesson 13：Read and write numbers within 1,000 after modeling with place value disks.
- Lesson 14：Model numbers with more than 9 ones or 9 tens; write in expanded, unit, standard, and word forms.
- Lesson 15：Explore a situation with more than 9 groups of ten.
- 海报要点：满十换一

F主题 比较三位数（Topic F: Comparing Two Three-Digit Numbers）

- Lesson 16：Compare two three-digit numbers using <, >, and =.
- Lesson 17：Compare two three-digit numbers using <, >, and = when there are more than 9 ones or 9 tens.

- Lesson 18：Order numbers in different forms. (Optional)
- 海报要点：用 <、>、= 比较

G主题 多 1、10、100 与少 1、10、100（Topic G: Finding 1, 10, and 100 More or Less Than a Number）

- Lesson 19：Model and use language to tell about 1 more and 1 less, 10 more and 10 less, and 100 more and 100 less.
- Lesson 20：Model 1 more and 1 less, 10 more and 10 less, and 100 more and 100 less when changing the hundreds place.
- Lesson 21：Complete a pattern counting up and down.
- 海报要点：数位上的变化

第4模块 200 以内加减法与 100 以内应用题（Module 4: Addition and Subtraction Within 200 with Word Problems to 100）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 100 以内的加减（Topic A: Sums and Differences Within 100）

- Lesson 1：Relate 1 more, 1 less, 10 more, and 10 less to addition and subtraction of 1 and 10.
- Lesson 2：Add and subtract multiples of 10 including counting on to subtract.
- Lesson 3：Add and subtract multiples of 10 and some ones within 100.
- Lesson 4：Add and subtract multiples of 10 and some ones within 100.
- Lesson 5：Solve one- and two-step word problems within 100 using strategies based on place value.
- 海报要点：多 10、少 10；加减整十

B主题 组成一个十（Topic B: Strategies for Composing a Ten）

- Lesson 6：Use manipulatives to represent the composition of 10 ones as 1 ten with two-digit addends.
- Lesson 7：Relate addition using manipulatives to a written vertical method.
- Lesson 8：Use math drawings to represent the composition and relate drawings to a written method.
- Lesson 9：Use math drawings to represent the composition when adding a two-digit to a three-digit addend.
- Lesson 10：Use math drawings to represent the composition when adding a two-digit to a three-digit addend.
- 海报要点：满十进一；竖式加法

C主题 分解一个十（Topic C: Strategies for Decomposing a Ten）

- Lesson 11：Represent subtraction with and without the decomposition of 1 ten as 10 ones with manipulatives.
- Lesson 12：Relate manipulative representations to a written method.
- Lesson 13：Use math drawings to represent subtraction with and without decomposition and relate drawings to a written method.
- Lesson 14：Represent subtraction with and without the decomposition when there is a three-digit minuend.

- Lesson 15：Represent subtraction with and without the decomposition when there is a three-digit minuend.
- Lesson 16：Solve one- and two-step word problems within 100 using strategies based on place value.
- 海报要点：退位：1 个十换 10 个一

D主题 组成十和百（Topic D: Strategies for Composing Tens and Hundreds）

- Lesson 17：Use mental strategies to relate compositions of 10 tens as 1 hundred to 10 ones as 1 ten.
- Lesson 18：Use manipulatives to represent additions with two compositions.
- Lesson 19：Relate manipulative representations to a written method.
- Lesson 20：Use math drawings to represent additions with up to two compositions and relate drawings to a written method.
- Lesson 21：Use math drawings to represent additions with up to two compositions and relate drawings to a written method.
- Lesson 22：Solve additions with up to four addends with totals within 200 with and without two compositions of larger units.
- 海报要点：两次进位

E主题 分解十和百（Topic E: Strategies for Decomposing Tens and Hundreds）

- Lesson 23：Use number bonds to break apart three-digit minuends and subtract from the hundred.
- Lesson 24：Use manipulatives to represent subtraction with decompositions of 1 hundred as 10 tens and 1 ten as 10 ones.
- Lesson 25：Relate manipulative representations to a written method.
- Lesson 26：Use math drawings to represent subtraction with up to two decompositions and relate drawings to a written method.
- Lesson 27：Subtract from 200 and from numbers with zeros in the tens place.
- Lesson 28：Subtract from 200 and from numbers with zeros in the tens place.
- 海报要点：两次退位；从 200 里减

F主题 书写方法的解释（Topic F: Student Explanations of Written Methods）

- Lesson 29：Use and explain the totals below method using words, math drawings, and numbers.
- Lesson 30：Compare totals below to new groups below as written methods.
- Lesson 31：Solve two-step word problems within 100.
- 海报要点：两种竖式记录方法

第5模块 1000 以内加减法与 100 以内应用题（Module 5: Addition and Subtraction Within 1,000 with Word Problems to 100）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 1000 以内加减策略（Topic A: Strategies for Adding and Subtracting Within 1,000）

- Lesson 1：Relate 10 more, 10 less, 100 more, and 100 less to addition and subtraction of 10 and 100.
- Lesson 2：Add and subtract multiples of 100, including counting on to subtract.
- Lesson 3：Add multiples of 100 and some tens within 1,000.
- Lesson 4：Subtract multiples of 100 and some tens within 1,000.
- Lesson 5：Use the associative property to make a hundred in one addend.

- Lesson 6：Use the associative property to subtract from three-digit numbers and verify solutions with addition.
- Lesson 7：Share and critique solution strategies for varied addition and subtraction problems within 1,000.
- 海报要点：加减整百；凑整百

B主题 1000 以内组成十和百（Topic B: Strategies for Composing Tens and Hundreds Within 1,000）

- Lesson 8：Relate manipulative representations to the addition algorithm.
- Lesson 9：Relate manipulative representations to the addition algorithm.
- Lesson 10：Use math drawings to represent additions with up to two compositions and relate drawings to the addition algorithm.
- Lesson 11：Use math drawings to represent additions with up to two compositions and relate drawings to the addition algorithm.
- Lesson 12：Choose and explain solution strategies and record with a written addition method.
- 海报要点：进位加法竖式

C主题 1000 以内分解十和百（Topic C: Strategies for Decomposing Tens and Hundreds Within 1,000）

- Lesson 13：Relate manipulative representations to the subtraction algorithm, and use addition to explain why the subtraction method works.
- Lesson 14：Use math drawings to represent subtraction with up to two decompositions, relate drawings to the algorithm, and use addition to explain why the subtraction method works.
- Lesson 15：Use math drawings to represent subtraction with up to two decompositions, relate drawings to the algorithm, and use addition to explain why the subtraction method works.
- Lesson 16：Subtract from multiples of 100 and from numbers with zero in the tens place.
- Lesson 17：Subtract from multiples of 100 and from numbers with zero in the tens place.
- Lesson 18：Apply and explain alternate methods for subtracting from multiples of 100 and from numbers with zero in the tens place.
- 海报要点：退位减法；中间有 0

D主题 选择解题方法（Topic D: Student Explanations for Choice of Solution Methods）

- Lesson 19：Choose and explain solution strategies and record with a written addition or subtraction method.
- Lesson 20：Choose and explain solution strategies and record with a written addition or subtraction method.
- 海报要点：说明为什么选这种方法

第6模块 乘法和除法的基础（Module 6: Foundations of Multiplication and Division）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 相等的组（Topic A: Formation of Equal Groups）

- Lesson 1：Use manipulatives to create equal groups.
- Lesson 2：Use math drawings to represent equal groups, and relate to repeated addition.
- Lesson 3：Use math drawings to represent equal groups, and relate to repeated addition.
- Lesson 4：Represent equal groups with tape diagrams, and relate to repeated addition.
- 海报要点：每组同样多；连加

B主题 阵列与相等的组（Topic B: Arrays and Equal Groups）

- Lesson 5：Compose arrays from rows and columns, and count to find the total using objects.
- Lesson 6：Decompose arrays into rows and columns, and relate to repeated addition.
- Lesson 7：Represent arrays and distinguish rows and columns using math drawings.
- Lesson 8：Create arrays using square tiles with gaps.
- Lesson 9：Solve word problems involving addition of equal groups in rows and columns.
- 海报要点：行与列；按行连加

C主题 长方形阵列（Topic C: Rectangular Arrays as a Foundation for Multiplication and Division）

- Lesson 10：Use square tiles to compose a rectangle, and relate to the array model.
- Lesson 11：Use square tiles to compose a rectangle, and relate to the array model.
- Lesson 12：Use math drawings to compose a rectangle with square tiles.
- Lesson 13：Use square tiles to decompose a rectangle.
- Lesson 14：Use scissors to partition a rectangle into same-size squares, and compose arrays with the squares.
- Lesson 15：Use math drawings to partition a rectangle with square tiles, and relate to repeated addition.
- Lesson 16：Use grid paper to create designs to develop spatial structuring.
- 海报要点：用正方形瓷砖拼长方形

D主题 奇数和偶数（Topic D: The Meaning of Even and Odd Numbers）

- Lesson 17：Relate doubles to even numbers, and write number sentences to express the sums.
- Lesson 18：Pair objects and skip-count to relate to even numbers.
- Lesson 19：Investigate the pattern of even numbers: 0, 2, 4, 6, and 8 in the ones place, and relate to odd numbers.
- Lesson 20：Use rectangular arrays to investigate odd and even numbers.
- 海报要点：两两配对；个位是 0、2、4、6、8

第7模块 长度、人民币和数据的问题解决（Module 7: Problem Solving with Length, Money, and Data）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 分类数据（Topic A: Problem Solving with Categorical Data）

- Lesson 1：Sort and record data into a table using up to four categories; use category counts to solve word problems.
- Lesson 2：Draw and label a picture graph to represent data with up to four categories.
- Lesson 3：Draw and label a bar graph to represent data; relate the count scale to the number line.
- Lesson 4：Draw a bar graph to represent a given data set.
- Lesson 5：Solve word problems using data presented in a bar graph.
- 海报要点：统计表、象形统计图、条形统计图

B主题 硬币和纸币（Topic B: Problem Solving with Coins and Bills）

- Lesson 6：Recognize the value of coins and count up to find their total value.
- Lesson 7：Solve word problems involving the total value of a group of coins.
- Lesson 8：Solve word problems involving the total value of a group of bills.

- Lesson 9：Solve word problems involving different combinations of coins with the same total value.
- Lesson 10：Use the fewest number of coins to make a given value.
- Lesson 11：Use different strategies to make $1 or make change from $1.
- Lesson 12：Solve word problems involving different ways to make change from $1.
- Lesson 13：Solve two-step word problems involving dollars or cents with totals within $100 or $1.
- 海报要点：人民币：元、角；找零
- 本地化：原书为美元、美分，本海报改为人民币（元、角）。

C主题 自制尺子（Topic C: Creating an Inch Ruler）

- Lesson 14：Connect measurement with physical units by using iteration with an inch tile to measure.
- Lesson 15：Apply concepts to create inch rulers; measure lengths using inch rulers.
- 海报要点：用 1 厘米方块做尺子
- 本地化：原书用 1 英寸方块做英寸尺，本海报改用 1 厘米方块做厘米尺。

D主题 测量和估计长度（Topic D: Measuring and Estimating Length Using Customary and Metric Units）

- Lesson 16：Measure various objects using inch rulers and yardsticks.
- Lesson 17：Develop estimation strategies by applying prior knowledge of length and using mental benchmarks.
- Lesson 18：Measure an object twice using different length units and compare; relate measurement to unit size.
- Lesson 19：Measure to compare the differences in lengths using inches, feet, and yards.
- 海报要点：厘米、分米、米
- 本地化：原书同时用英寸、英尺、码和厘米、米，本海报只用公制（厘米、分米、米）。

E主题 长度应用题（Topic E: Problem Solving with Customary and Metric Units）

- Lesson 20：Solve two-digit addition and subtraction word problems involving length by using tape diagrams and writing equations to represent the problem.
- Lesson 21：Identify unknown numbers on a number line diagram by using the distance between numbers and reference points.
- Lesson 22：Represent two-digit sums and differences involving length by using the ruler as a number line.
- 海报要点：用尺子当数线
- 本地化：原书含英制长度应用题，本海报只用公制。

F主题 测量数据（Topic F: Displaying Measurement Data）

- Lesson 23：Collect and record measurement data in a table; answer questions and summarize the data set.
- Lesson 24：Draw a line plot to represent the measurement data; relate the measurement scale to the number line.
- Lesson 25：Draw a line plot to represent a given data set; answer questions and draw conclusions based on measurement data.
- Lesson 26：Draw a line plot to represent a given data set; answer questions and draw conclusions based on measurement data.

- 海报要点：线图表示测量数据
- 本地化：原书线图数据以英寸为单位，本海报改为厘米。

第8模块 时间、图形和分数（Module 8: Time, Shapes, and Fractions as Equal Parts of Shapes）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 几何图形的特征（Topic A: Attributes of Geometric Shapes）

- Lesson 1：Describe two-dimensional shapes based on attributes.
- Lesson 2：Build, identify, and analyze two-dimensional shapes with specified attributes.
- Lesson 3：Use attributes to draw different polygons including triangles, quadrilaterals, pentagons, and hexagons.
- Lesson 4：Use attributes to identify and draw different quadrilaterals including rectangles, rhombuses, parallelograms, and trapezoids.
- Lesson 5：Relate the square to the cube, and describe the cube based on attributes.
- 海报要点：边数、角数；三角形、四边形、五边形、六边形

B主题 组合图形与分数概念（Topic B: Composite Shapes and Fraction Concepts）

- Lesson 6：Combine shapes to create a composite shape; create a new shape from composite shapes.
- Lesson 7：Interpret equal shares in composite shapes as halves, thirds, and fourths.
- Lesson 8：Interpret equal shares in composite shapes as halves, thirds, and fourths.
- 海报要点：拼图形；二分之一、三分之一、四分之一

C主题 圆和长方形的等分（Topic C: Halves, Thirds, and Fourths of Circles and Rectangles）

- Lesson 9：Partition circles and rectangles into equal parts, and describe those parts as halves, thirds, or fourths.
- Lesson 10：Partition circles and rectangles into equal parts, and describe those parts as halves, thirds, or fourths.
- Lesson 11：Describe a whole by the number of equal parts including 2 halves, 3 thirds, and 4 fourths.
- Lesson 12：Recognize that equal parts of an identical rectangle can have different shapes.
- 海报要点：平均分成 2、3、4 份

D主题 用分数认识时间（Topic D: Application of Fractions to Tell Time）

- Lesson 13：Construct a paper clock by partitioning a circle into halves and quarters, and tell time to the half hour or quarter hour.
- Lesson 14：Tell time to the nearest five minutes.
- Lesson 15：Tell time to the nearest five minutes; relate a.m. and p.m. to time of day.
- Lesson 16：Solve elapsed time problems involving whole hours and a half hour.
- 海报要点：整时、半时、一刻；5 分钟

三年级（Grade 3）· 7 个模块 · 35 个 Topic

第1模块 乘除法的性质与 2–5、10 的问题（Module 1: Properties of Multiplication and Division and Solving Problems with Units of 2–5 and 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 乘法与因数的意义（Topic A: Multiplication and the Meaning of the Factors）

- Lesson 1：Understand equal groups of as multiplication.
- Lesson 2：Relate multiplication to the array model.
- Lesson 3：Interpret the meaning of factors—the size of the group or the number of groups.
- 海报要点：相等的组；阵列；因数

B主题 除法是求未知因数（Topic B: Division as an Unknown Factor Problem）

- Lesson 4：Understand the meaning of the unknown as the size of the group in division.
- Lesson 5：Understand the meaning of the unknown as the number of groups in division.
- Lesson 6：Interpret the unknown in division using the array model.
- 海报要点：平均分：每组几个、分几组

C主题 2 和 3 的乘法（Topic C: Multiplication Using Units of 2 and 3）

- Lesson 7：Demonstrate the commutativity of multiplication, and practice related facts by skip-counting objects in array models.
- Lesson 8：Demonstrate the commutativity of multiplication, and practice related facts by skip-counting objects in array models.
- Lesson 9：Find related multiplication facts by adding and subtracting equal groups in array models.
- Lesson 10：Model the distributive property with arrays to decompose units as a strategy to multiply.
- 海报要点：交换律；分配律

D主题 2 和 3 的除法（Topic D: Division Using Units of 2 and 3）

- Lesson 11：Model division as the unknown factor in multiplication using arrays and tape diagrams.
- Lesson 12：Interpret the quotient as the number of groups or the number of objects in each group using units of 2.
- Lesson 13：Interpret the quotient as the number of groups or the number of objects in each group using units of 3.
- 海报要点：除法和乘法的关系

E主题 4 的乘法和除法（Topic E: Multiplication and Division Using Units of 4）

- Lesson 14：Skip-count objects in models to build fluency with multiplication facts using units of 4.
- Lesson 15：Relate arrays to tape diagrams to model the commutative property of multiplication.
- Lesson 16：Use the distributive property as a strategy to find related multiplication facts.
- Lesson 17：Model the relationship between multiplication and division.
- 海报要点：数线上数 4；阵列

F主题 分配律与解决问题（Topic F: Distributive Property and Problem Solving Using Units of 2–5 and 10）

- Lesson 18：Apply the distributive property to decompose units.
- Lesson 19：Apply the distributive property to decompose units.
- Lesson 20：Solve two-step word problems involving multiplication and division, and assess the reasonableness of answers.
- Lesson 21：Solve two-step word problems involving all four operations, and assess the reasonableness of answers.
- 海报要点：拆成 5 和几；两步问题

第2模块 位值与测量单位的问题（Module 2: Place Value and Problem Solving with Units of Measure）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 时间（Topic A: Time Measurement and Problem Solving）

- Lesson 1：Explore time as a continuous measurement using a stopwatch.
- Lesson 2：Relate skip-counting by fives on the clock and telling time to a continuous measurement model, the number line.
- Lesson 3：Count by fives and ones on the number line as a strategy to tell time to the nearest minute on the clock.
- Lesson 4：Solve word problems involving time intervals within 1 hour by counting backward and forward using the number line and clock.
- Lesson 5：Solve word problems involving time intervals within 1 hour by adding and subtracting on the number line.
- 海报要点：精确到分；经过时间

B主题 公制的重量和容量（Topic B: Measuring Weight and Liquid Volume in Metric Units）

- Lesson 6：Build and decompose a kilogram to reason about the size and weight of 1 kilogram, 100 grams, 10 grams, and 1 gram.
- Lesson 7：Develop estimation strategies by reasoning about the weight in kilograms of a series of familiar objects to establish mental benchmark measures.
- Lesson 8：Solve one-step word problems involving metric weights within 100 and estimate to reason about solutions.
- Lesson 9：Decompose a liter to reason about the size of 1 liter, 100 milliliters, 10 milliliters, and 1 milliliter.
- Lesson 10：Estimate and measure liquid volume in liters and milliliters using the vertical number line.
- Lesson 11：Solve mixed word problems involving all four operations with grams, kilograms, liters, and milliliters given in the same units.
- 海报要点：千克、克；升、毫升

C主题 近似到十和百（Topic C: Rounding to the Nearest Ten and Hundred）

- Lesson 12：Round two-digit measurements to the nearest ten on the vertical number line.
- Lesson 13：Round two- and three-digit numbers to the nearest ten on the vertical number line.
- Lesson 14：Round to the nearest hundred on the vertical number line.

- 海报要点：竖直数线；四舍五入

D主题 测量中的加法竖式（Topic D: Two- and Three-Digit Measurement Addition Using the Standard Algorithm）

- Lesson 15：Add measurements using the standard algorithm to compose larger units once.
- Lesson 16：Add measurements using the standard algorithm to compose larger units twice.
- Lesson 17：Estimate sums by rounding and apply to solve measurement word problems.
- 海报要点：两三位数加法

E主题 测量中的减法竖式（Topic E: Two- and Three-Digit Measurement Subtraction Using the Standard Algorithm）

- Lesson 18：Decompose once to subtract measurements including three-digit minuends with zeros in the tens or ones place.
- Lesson 19：Decompose twice to subtract measurements including three- digit minuends with zeros in the tens and ones places.
- Lesson 20：Estimate differences by rounding and apply to solve measurement word problems.
- Lesson 21：Estimate sums and differences of measurements by rounding, and then solve mixed word problems.
- 海报要点：两三位数减法

第3模块 0、1、6–9 和整十的乘除法（Module 3: Multiplication and Division with Units of 0, 1, 6–9, and Multiples of 10）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 乘除法的性质（Topic A: The Properties of Multiplication and Division）

- Lesson 1：Study commutativity to find known facts of 6, 7, 8, and 9.
- Lesson 2：Apply the distributive and commutative properties to relate multiplication facts $5 \times n + n$ to $6 \times n$ and $n \times 6$ where n is the size of the unit.
- Lesson 3：Multiply and divide with familiar facts using a letter to represent the unknown.
- 海报要点：交换律；乘除互逆

B主题 6 和 7 的乘除（Topic B: Multiplication and Division Using Units of 6 and 7）

- Lesson 4：Count by units of 6 to multiply and divide using number bonds to decompose.
- Lesson 5：Count by units of 7 to multiply and divide using number bonds to decompose.
- Lesson 6：Use the distributive property as a strategy to multiply and divide using units of 6 and 7.
- Lesson 7：Interpret the unknown in multiplication and division to model and solve problems using units of 6 and 7.
- 海报要点：分配律：$6 \times 7 = 5 \times 7 + 1 \times 7$

C主题 8 以内的乘除（Topic C: Multiplication and Division Using Units up to 8）

- Lesson 8：Understand the function of parentheses and apply to solving problems.
- Lesson 9：Model the associative property as a strategy to multiply.
- Lesson 10：Use the distributive property as a strategy to multiply and divide.
- Lesson 11：Interpret the unknown in multiplication and division to model and solve problems.
- 海报要点：括号；结合律

D主题 9 的乘除（Topic D: Multiplication and Division Using Units of 9）

- Lesson 12：Apply the distributive property and the fact 9 = 10 – 1 as a strategy to multiply.
- Lesson 13：Identify and use arithmetic patterns to multiply.
- Lesson 14：Identify and use arithmetic patterns to multiply.
- Lesson 15：Interpret the unknown in multiplication and division to model and solve problems.
- 海报要点：9 = 10 − 1；规律

E主题 规律、0 和 1（Topic E: Analysis of Patterns and Problem Solving Including Units of 0 and 1）

- Lesson 16：Reason about and explain arithmetic patterns using units of 0 and 1 as they relate to multiplication and division.
- Lesson 17：Identify patterns in multiplication and division facts using the multiplication table.
- Lesson 18：Solve two-step word problems involving all four operations and assess the reasonableness of solutions.
- 海报要点：乘 0、乘 1；乘法表规律

F主题 一位数乘整十数（Topic F: Multiplication of Single-Digit Factors and Multiples of 10）

- Lesson 19：Multiply by multiples of 10 using the place value chart.
- Lesson 20：Use place value strategies and the associative property n × (m × 10) = (n × m) × 10 (where n and m are less than 10) to multiply by multiples of 10.
- Lesson 21：Solve two-step word problems involving multiplying single-digit factors and multiples of 10.
- 海报要点：3 × 40 = 3 × 4 个十

第4模块 乘法与面积（Module 4: Multiplication and Area）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 认识面积（Topic A: Foundations for Understanding Area）

- Lesson 1：Understand area as an attribute of plane figures.
- Lesson 2：Decompose and recompose shapes to compare areas.
- Lesson 3：Model tiling with centimeter and inch unit squares as a strategy to measure area.
- Lesson 4：Relate side lengths with the number of tiles on a side.
- 海报要点：单位正方形铺满

B主题 面积的测量（Topic B: Concepts of Area Measurement）

- Lesson 5：Form rectangles by tiling with unit squares to make arrays.
- Lesson 6：Draw rows and columns to determine the area of a rectangle given an incomplete array.
- Lesson 7：Interpret area models to form rectangular arrays.
- Lesson 8：Find the area of a rectangle through multiplication of the side lengths.
- 海报要点：平方厘米、平方米；数格子

C主题 面积模型与运算律（Topic C: Arithmetic Properties Using Area Models）

- Lesson 9：Analyze different rectangles and reason about their area.
- Lesson 10：Apply the distributive property as a strategy to find the total area of a large rectangle by adding two products.
- Lesson 11：Demonstrate the possible whole number side lengths of rectangles with areas of 24, 36, 48, or 72 square units using the associative property.

- 海报要点：分配律：拆长方形

D主题 面积的应用（Topic D: Applications of Area Using Side Lengths of Figures）

- Lesson 12：Solve word problems involving area.
- Lesson 13：Find areas by decomposing into rectangles or completing composite figures to form rectangles.
- Lesson 14：Find areas by decomposing into rectangles or completing composite figures to form rectangles.
- Lesson 15：Apply knowledge of area to determine areas of rooms in a given floor plan.
- Lesson 16：Apply knowledge of area to determine areas of rooms in a given floor plan.
- 海报要点：组合图形面积

第5模块 数线上的分数（Module 5: Fractions as Numbers on the Number Line）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 把整体平均分（Topic A: Partitioning a Whole into Equal Parts）

- Lesson 1：Specify and partition a whole into equal parts, identifying and counting unit fractions using concrete models.
- Lesson 2：Specify and partition a whole into equal parts, identifying and counting unit fractions by folding fraction strips.
- Lesson 3：Specify and partition a whole into equal parts, identifying and counting unit fractions by drawing pictorial area models.
- Lesson 4：Represent and identify fractional parts of different wholes.
- 海报要点：等分；份数

B主题 单位分数和整体（Topic B: Unit Fractions and Their Relation to the Whole）

- Lesson 5：Partition a whole into equal parts and define the equal parts to identify the unit fraction numerically.
- Lesson 6：Build non-unit fractions less than one whole from unit fractions.
- Lesson 7：Identify and represent shaded and non-shaded parts of one whole as fractions.
- Lesson 8：Represent parts of one whole as fractions with number bonds.
- Lesson 9：Build and write fractions greater than one whole using unit fractions.
- 海报要点：几个 1/b 组成整体

C主题 比较单位分数（Topic C: Comparing Unit Fractions and Specifying the Whole）

- Lesson 10：Compare unit fractions by reasoning about their size using fraction strips.
- Lesson 11：Compare unit fractions with different-sized models representing the whole.
- Lesson 12：Specify the corresponding whole when presented with one equal part.
- Lesson 13：Identify a shaded fractional part in different ways depending on the designation of the whole.
- 海报要点：分母越大，每份越小

D主题 数线上的分数（Topic D: Fractions on the Number Line）

- Lesson 14：Place fractions on a number line with endpoints 0 and 1.
- Lesson 15：Place any fraction on a number line with endpoints 0 and 1.

- Lesson 16：Place whole number fractions and fractions between whole numbers on the number line.
- Lesson 17：Practice placing various fractions on the number line.
- Lesson 18：Compare fractions and whole numbers on the number line by reasoning about their distance from 0.
- Lesson 19：Understand distance and position on the number line as strategies for comparing fractions. (Optional)
- 海报要点：0 到 1 平均分

E主题 相等的分数（Topic E: Equivalent Fractions）

- Lesson 20：Recognize and show that equivalent fractions have the same size, though not necessarily the same shape.
- Lesson 21：Recognize and show that equivalent fractions refer to the same point on the number line.
- Lesson 22：Generate simple equivalent fractions by using visual fraction models and the number line.
- Lesson 23：Generate simple equivalent fractions by using visual fraction models and the number line.
- Lesson 24：Express whole numbers as fractions and recognize equivalence with different units.
- Lesson 25：Express whole number fractions on the number line when the unit interval is 1.
- Lesson 26：Decompose whole number fractions greater than 1 using whole number equivalence with various models.
- Lesson 27：Explain equivalence by manipulating units and reasoning about their size.
- 海报要点：1/2 = 2/4；整数写成分数

F主题 分数的比较和排序（Topic F: Comparison, Order, and Size of Fractions）

- Lesson 28：Compare fractions with the same numerator pictorially.
- Lesson 29：Compare fractions with the same numerator using <, >, or =, and use a model to reason about their size.
- Lesson 30：Partition various wholes precisely into equal parts using a number line method.
- 海报要点：同分子、同分母比较

第6模块 收集和整理数据（Module 6: Collecting and Displaying Data）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 分类数据（Topic A: Generate and Analyze Categorical Data）

- Lesson 1：Generate and organize data.
- Lesson 2：Rotate tape diagrams vertically.
- Lesson 3：Create scaled bar graphs.
- Lesson 4：Solve one- and two-step problems involving graphs.
- 海报要点：比例条形图；象形统计图

B主题 测量数据（Topic B: Generate and Analyze Measurement Data）

- Lesson 5：Create ruler with 1-inch, -inch, and -inch intervals, and 2 4 generate measurement data.
- Lesson 6：Interpret measurement data from various line plots.

- Lesson 7：Represent measurement data with line plots.
- Lesson 8：Represent measurement data with line plots.
- Lesson 9：Analyze data to problem solve.
- 海报要点：用线图表示测量数据
- 本地化：原书用 1/2、1/4 英寸刻度的尺子收集数据，本海报改为厘米和半厘米。

第7模块 几何与测量应用题（Module 7: Geometry and Measurement Word Problems）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 解决应用题（Topic A: Solving Word Problems）

- Lesson 1：Solve word problems in varied contexts using a letter to represent the unknown.
- Lesson 2：Solve word problems in varied contexts using a letter to represent the unknown.
- Lesson 3：Share and critique peer solution strategies to varied word problems.
- 海报要点：四则运算；画图

B主题 平面图形的特征（Topic B: Attributes of Two-Dimensional Figures）

- Lesson 4：Compare and classify quadrilaterals.
- Lesson 5：Compare and classify other polygons.
- Lesson 6：Draw polygons with specified attributes to solve problems.
- Lesson 7：Reason about composing and decomposing polygons using tetrominoes.
- Lesson 8：Create a tangram puzzle and observe relationships among the shapes.
- Lesson 9：Reason about composing and decomposing polygons using tangrams.
- 海报要点：四边形；多边形

C主题 周长（Topic C: Problem Solving with Perimeter）

- Lesson 10：Decompose quadrilaterals to understand perimeter as the boundary of a shape.
- Lesson 11：Tessellate to understand perimeter as the boundary of a shape. (Optional.)
- Lesson 12：Measure side lengths in whole number units to determine the perimeter of polygons.
- Lesson 13：Explore perimeter as an attribute of plane figures and solve problems.
- Lesson 14：Determine the perimeter of regular polygons and rectangles when whole number measurements are unknown.
- Lesson 15：Solve word problems to determine perimeter with given side lengths.
- Lesson 16：Use string to measure the perimeter of various circles to the nearest quarter inch.
- Lesson 17：Use all four operations to solve problems involving perimeter and unknown measurements.
- 海报要点：围一圈的长度

D主题 周长和面积的线图（Topic D: Recording Perimeter and Area Data on Line Plots）

- Lesson 18：Construct rectangles from a given number of unit squares and determine the perimeters.
- Lesson 19：Use a line plot to record the number of rectangles constructed from a given number of unit squares.
- Lesson 20：Construct rectangles with a given perimeter using unit squares and determine their areas.
- Lesson 21：Construct rectangles with a given perimeter using unit squares and determine their areas.

- Lesson 22：Use a line plot to record the number of rectangles constructed in Lessons 20 and 21.
- 海报要点：面积相同、周长不同

E主题 周长与面积的问题（Topic E: Problem Solving with Perimeter and Area）

- Lesson 23：Solve a variety of word problems with perimeter.
- Lesson 24：Use rectangles to draw a robot with specified perimeter measurements, and reason about the different areas that may be produced.
- Lesson 25：Use rectangles to draw a robot with specified perimeter measurements, and reason about the different areas that may be produced.
- Lesson 26：Use rectangles to draw a robot with specified perimeter measurements, and reason about the different areas that may be produced.
- Lesson 27：Use rectangles to draw a robot with specified perimeter measurements, and reason about the different areas that may be produced.
- Lesson 28：Solve a variety of word problems involving area and perimeter using all four operations.
- Lesson 29：Solve a variety of word problems involving area and perimeter using all four operations.
- Lesson 30：Share and critique peer strategies for problem solving.
- 海报要点：设计；比较

F主题 学年回顾（Topic F: Year in Review）

- Lesson 31：Explore and create unconventional representations of one-half.
- Lesson 32：Explore and create unconventional representations of one-half.
- Lesson 33：Solidify fluency with Grade 3 skills.
- Lesson 34：Create resource booklets to support fluency with Grade 3 skills.
- 海报要点：游戏复习乘法、分数

四年级（Grade 4）· 7 个模块 · 37 个 Topic

第1模块 位值、近似与加减法算法（Module 1: Place Value, Rounding, and Algorithms for Addition and Subtraction）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 多位数的位值（Topic A: Place Value of Multi-Digit Whole Numbers）

- Lesson 1：Interpret a multiplication equation as a comparison.
- Lesson 2：Recognize a digit represents 10 times the value of what it represents in the place to its right.
- Lesson 3：Name numbers within 1 million by building understanding of the place value chart and placement of commas for naming base thousand units.
- Lesson 4：Read and write multi-digit numbers using base ten numerals, number names, and expanded form.
- 海报要点：每一位是右边一位的 10 倍

B主题 比较多位数（Topic B: Comparing Multi-Digit Whole Numbers）

- Lesson 5：Compare numbers based on meanings of the digits using >, <, or = to record the comparison.
- Lesson 6：Find 1, 10, and 100 thousand more and less than a given number.
- 海报要点：从高位比起

C主题 多位数的近似（Topic C: Rounding Multi-Digit Whole Numbers）

- Lesson 7：Round multi-digit numbers to the thousands place using the vertical number line.
- Lesson 8：Round multi-digit numbers to any place using the vertical number line.
- Lesson 9：Use place value understanding to round multi-digit numbers to any place value.
- Lesson 10：Use place value understanding to round multi-digit numbers to any place value using real world applications.
- 海报要点：近似到任何一位

D主题 多位数加法（Topic D: Multi-Digit Whole Number Addition）

- Lesson 11：Use place value understanding to fluently add multi-digit whole numbers using the standard addition algorithm, and apply the algorithm to solve word problems using tape diagrams.
- Lesson 12：Solve multi-step word problems using the standard addition algorithm modeled with tape diagrams, and assess the reasonableness of answers using rounding.
- 海报要点：标准竖式

E主题 多位数减法（Topic E: Multi-Digit Whole Number Subtraction）

- Lesson 13：Use place value understanding to decompose to smaller units once using the standard subtraction algorithm, and apply the algorithm to solve word problems using tape diagrams.
- Lesson 14：Use place value understanding to decompose to smaller units up to three times using the standard subtraction algorithm, and apply the algorithm to solve word problems using tape diagrams.

- Lesson 15：Use place value understanding to fluently decompose to smaller units multiple times in any place using the standard subtraction algorithm, and apply the algorithm to solve word problems using tape diagrams.
- Lesson 16：Solve two-step word problems using the standard subtraction algorithm fluently modeled with tape diagrams, and assess the reasonableness of answers using rounding.
- 海报要点：连续退位

F主题 加减应用题（Topic F: Addition and Subtraction Word Problems）

- Lesson 17：Solve additive compare word problems modeled with tape diagrams.
- Lesson 18：Solve multi-step word problems modeled with tape diagrams, and assess the reasonableness of answers using rounding.
- Lesson 19：Create and solve multi-step word problems from given tape diagrams and equations.
- 海报要点：条形图；多步问题

第2模块 公制单位换算与解决问题（Module 2: Unit Conversions and Problem Solving with Metric Measurement）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 公制单位换算（Topic A: Metric Unit Conversions）

- Lesson 1：Express metric length measurements in te rms of a smaller unit; model and solve addition and subtraction word problems involving metric length.
- Lesson 2：Express metric mass measurements in terms of a smaller unit; model and solve addition and subtraction word problems involving metric mass.
- Lesson 3：Express metric capacity mea surements in terms of a smaller unit; model and solve addition and subtraction word problems involving metric capacity.
- 海报要点：千米、米、厘米；千克、克；升、毫升

B主题 单位换算的应用（Topic B: Application of Metric Unit Conversions）

- Lesson 4：Know and relate metric units to place value units in order to express measurements in different units.
- Lesson 5：Use addition and subtraction to solve multi-step word problems involving length, mass, and capacity.
- 海报要点：混合单位加减

第3模块 多位数乘除法（Module 3: Multi-Digit Multiplication and Division）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 倍数比较应用题（Topic A: Multiplicative Comparison Word Problems）

- Lesson 1：Investigate and use the formulas for area and perimeter of rectangles.
- Lesson 2：Solve multiplicative comparison word problems by applying the area and perimeter formulas.
- Lesson 3：Demonstrate understanding of area and perimeter formulas by solving multi-step real-world problems.
- 海报要点：是几倍；周长和面积

B主题 乘 10、100、1000（Topic B: Multiplication by 10, 100, and 1,000）

- Lesson 4：Interpret and represent patterns when multiplying by 10, 100, and 1,000 in arrays and numerically.
- Lesson 5：Multiply multiples of 10, 100, and 1,000 by single digits, recognizing patterns.
- Lesson 6：Multiply two-digit multiples of 10 by two-digit multiples of 10 with the area model.
- 海报要点：数位左移

C主题 四位数乘一位数（Topic C: Multiplication of up to Four Digits by Single-Digit Numbers）

- Lesson 7：Use place value disks to represent two-digit by one-digit multiplication.
- Lesson 8：Extend the use of place value disks to represent three- and four- digit by one-digit multiplication.
- Lesson 9：Multiply three- and four-digit numbers by one-digit numbers applying the standard algorithm.
- Lesson 10：Multiply three- and four-digit numbers by one-digit numbers applying the standard algorithm.
- Lesson 11：Connect the area model and the partial products method to the standard algorithm.
- 海报要点：面积模型；部分积

D主题 乘法应用题（Topic D: Multiplication Word Problems）

- Lesson 12：Solve two-step word problems, including multiplicative comparison.
- Lesson 13：Use multiplication, addition, or subtraction to solve multi-step word problems.
- 海报要点：多步问题

E主题 两位数除以一位数（Topic E: Division of Tens and Ones with Successive Remainders）

- Lesson 14：Solve division word problems with remainders.
- Lesson 15：Understand and solve division problems with a remainder using the array and area models.
- Lesson 16：Understand and solve two-digit dividend division problems with a remainder in the ones place by using place value disks.
- Lesson 17：Represent and solve division problems requiring decomposing a remainder in the tens.
- Lesson 18：Find whole number quotients and remainders.
- Lesson 19：Explain remainders by using place value understanding and models.
- Lesson 20：Solve division problems without remainders using the area model.
- Lesson 21：Solve division problems with remainders using the area model.
- 海报要点：余数；连续余数

F主题 整除的推理（Topic F: Reasoning with Divisibility）

- Lesson 22：Find factor pairs for numbers to 100, and use understanding of factors to define prime and composite.
- Lesson 23：Use division and the associative property to test for factors and observe patterns.
- Lesson 24：Determine if a whole number is a multiple of another number.
- Lesson 25：Explore properties of prime and composite numbers to 100 by using multiples.
- 海报要点：因数、倍数、质数、合数

G主题 四位数除以一位数（Topic G: Division of Thousands, Hundreds, Tens, and Ones）

- Lesson 26：Divide multiples of 10, 100, and 1,000 by single-digit numbers.
- Lesson 27：Represent and solve division problems with up to a three-digit dividend numerically and with place value disks requiring decomposing a remainder in the hundreds place.
- Lesson 28：Represent and solve three-digit dividend division with divisors of 2, 3, 4, and 5 numerically.
- Lesson 29：Represent numerically four-digit dividend division with divisors of 2, 3, 4, and 5, decomposing a remainder up to three times.
- Lesson 30：Solve division problems with a zero in the dividend or with a zero in the quotient.
- Lesson 31：Interpret division word problems as either number of groups unknown or group size unknown.
- Lesson 32：Interpret and find whole number quotients and remainders to solve one-step division word problems with larger divisors of 6, 7, 8, and 9.
- Lesson 33：Explain the connection of the area model of division to the long division algorithm for three- and four-digit dividends.
- 海报要点：长除法；面积模型

H主题 两位数乘两位数（Topic H: Multiplication of Two-Digit by Two-Digit Numbers）

- Lesson 34：Multiply two-digit multiples of 10 by two-digit numbers using a place value chart.
- Lesson 35：Multiply two-digit multiples of 10 by two-digit numbers using the area model.
- Lesson 36：Multiply two-digit by two-digit numbers using four partial products.
- Lesson 37：Transition from four partial products to the standard algorithm for two-digit by two-digit multiplication.
- Lesson 38：Transition from four partial products to the standard algorithm for two-digit by two-digit multiplication.
- 海报要点：面积模型；竖式

第4模块 角的度量与平面图形（Module 4: Angle Measure and Plane Figures）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 线和角（Topic A: Lines and Angles）

- Lesson 1：Identify and draw points, lines, line segments, rays, and angles. Recognize them in various contexts and familiar figures.
- Lesson 2：Use right angles to determine whether angles are equal to, greater than, or less than right angles. Draw right, obtuse, and acute angles.
- Lesson 3：Identify, define, and draw perpendicular lines.
- Lesson 4：Identify, define, and draw parallel lines.
- 海报要点：点、线、线段、射线；垂直、平行

B主题 角的度量（Topic B: Angle Measurement）

- Lesson 5：Use a circular protractor to understand a 1-degree angle as 360 of a turn. Explore benchmark angles using the protractor.
- Lesson 6：Use varied protractors to distinguish angle measure from length measurement.
- Lesson 7：Measure and draw angles. Sketch given angle measures, and verify with a protractor.
- Lesson 8：Identify and measure angles as turns and recognize them in various contexts.

- 海报要点：量角器；度

C主题 角度的加法（Topic C: Problem Solving with the Addition of Angle Measures）

- Lesson 9：Decompose angles using pattern blocks.
- Lesson 10：Use the addition of adjacent angle measures to solve problems using a symbol for the unknown angle measure.
- Lesson 11：Use the addition of adjacent angle measures to solve problems using a symbol for the unknown angle measure.
- 海报要点：相邻角相加；未知角

D主题 平面图形与对称（Topic D: Two-Dimensional Figures and Symmetry）

- Lesson 12：Recognize lines of symmetry for given two-dimensional figures. Identify line-symmetric figures, and draw lines of symmetry.
- Lesson 13：Analyze and classify triangles based on side length, angle measure, or both.
- Lesson 14：Define and construct triangles from given criteria. Explore symmetry in triangles.
- Lesson 15：Classify quadrilaterals based on parallel and perpendicular lines and the presence or absence of angles of a specified size.
- Lesson 16：Reason about attributes to construct quadrilaterals on square or triangular grid paper.
- 海报要点：三角形分类；对称轴

第5模块 分数的相等、排序与运算（Module 5: Fraction Equivalence, Ordering, and Operations）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 分解与分数相等（Topic A: Decomposition and Fraction Equivalence）

- Lesson 1：Decompose fractions as a sum of unit fractions using tape diagrams.
- Lesson 2：Decompose fractions as a sum of unit fractions using tape diagrams.
- Lesson 3：Decompose non-unit fractions and represent them as a whole number times a unit fraction using tape diagrams.
- Lesson 4：Decompose fractions into sums of smaller unit fractions using tape diagrams.
- Lesson 5：Decompose unit fractions using area models to show equivalence.
- Lesson 6：Decompose fractions using area models to show equivalence.
- 海报要点：3/5 = 1/5 + 2/5

B主题 乘除法与分数相等（Topic B: Fraction Equivalence Using Multiplication and Division）

- Lesson 7：Use the area model and multiplication to show the equivalence of two fractions.
- Lesson 8：Use the area model and multiplication to show the equivalence of two fractions.
- Lesson 9：Use the area model and division to show the equivalence of two fractions.
- Lesson 10：Use the area model and division to show the equivalence of two fractions.
- Lesson 11：Explain fraction equivalence using a tape diagram and the number line, and relate that to the use of multiplication and division.
- 海报要点：分子分母同乘同除

C主题 比较分数（Topic C: Fraction Comparison）

- Lesson 12：Reason using benchmarks to compare two fractions on the number line.
- Lesson 13：Reason using benchmarks to compare two fractions on the number line.

- Lesson 14：Find common units or number of units to compare two fractions.
- Lesson 15：Find common units or number of units to compare two fractions.
- 海报要点：用基准、通分比较

D主题 分数加减法（Topic D: Fraction Addition and Subtraction）

- Lesson 16：Use visual models to add and subtract two fractions with the same units.
- Lesson 17：Use visual models to add and subtract two fractions with the same units, including subtracting from one whole.
- Lesson 18：Add and subtract more than two fractions.
- Lesson 19：Solve word problems involving addition and subtraction of fractions.
- Lesson 20：Use visual models to add two fractions with related units using the denominators 2, 3, 4, 5, 6, 8, 10, and 12.
- Lesson 21：Use visual models to add two fractions with related units using the denominators 2, 3, 4, 5, 6, 8, 10, and 12.
- 海报要点：同分母；相关分母

E主题 大于 1 的分数（Topic E: Extending Fraction Equivalence to Fractions Greater Than 1）

- Lesson 22：Add a fraction less than 1 to, or subtract a fraction less than 1 from, a whole number using decomposition and visual models.
- Lesson 23：Add and multiply unit fractions to build fractions greater than 1 using visual models.
- Lesson 24：Decompose and compose fractions greater than 1 to express them in various forms.
- Lesson 25：Decompose and compose fractions greater than 1 to express them in various forms.
- Lesson 26：Compare fractions greater than 1 by reasoning using benchmark fractions.
- Lesson 27：Compare fractions greater than 1 by creating common numerators or denominators.
- Lesson 28：Solve word problems with line plots.
- 海报要点：假分数与带分数

F主题 分解法加减分数（Topic F: Addition and Subtraction of Fractions by Decomposition）

- Lesson 29：Estimate sums and differences using benchmark numbers.
- Lesson 30：Add a mixed number and a fraction.
- Lesson 31：Add mixed numbers.
- Lesson 32：Subtract a fraction from a mixed number.
- Lesson 33：Subtract a mixed number from a mixed number.
- Lesson 34：Subtract mixed numbers.
- 海报要点：带分数加减

G主题 分数乘整数（Topic G: Repeated Addition of Fractions as Multiplication）

- Lesson 35：Represent the multiplication of n times a/b as (n × a)/b using the associative property and visual models.
- Lesson 36：Represent the multiplication of n times a/b as (n × a)/b using the associative property and visual models.
- Lesson 37：Find the product of a whole number and a mixed number using the distributive property.
- Lesson 38：Find the product of a whole number and a mixed number using the distributive property.
- Lesson 39：Solve multiplicative comparison word problems involving fractions.

- Lesson 40：Solve word problems involving the multiplication of a whole number and a fraction including those involving line plots.
- 海报要点：连加 = 乘法

H主题 分数规律（Topic H: Exploring a Fraction Pattern）

- Lesson 41：Find and use a pattern to calculate the sum of all fractional parts between 0 and 1. Share and critique peer strategies.
- 海报要点：规律探究

第6模块 小数（Module 6: Decimal Fractions）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 认识十分之一（Topic A: Exploration of Tenths）

- Lesson 1：Use metric measurement to model the decomposition of one whole into tenths.
- Lesson 2：Use metric measurement and area models to represent tenths as fractions greater than 1 and decimal numbers.
- Lesson 3：Represent mixed numbers with units of tens, ones, and tenths with place value disks, on the number line, and in expanded form.
- 海报要点：0.1；米尺上的分米

B主题 十分之一和百分之一（Topic B: Tenths and Hundredths）

- Lesson 4：Use meters to model the decomposition of one whole into hundredths. Represent and count hundredths.
- Lesson 5：Model the equivalence of tenths and hundredths using the area model and place value disks.
- Lesson 6：Use the area model and number line to represent mixed numbers with units of ones, tenths, and hundredths in fraction and decimal forms.
- Lesson 7：Model mixed numbers with units of hundreds, tens, ones, tenths, and hundredths in expanded form and on the place value chart.
- Lesson 8：Use understanding of fraction equivalence to investigate decimal numbers on the place value chart expressed in different units.
- 海报要点：0.01；百格图

C主题 比较小数（Topic C: Decimal Comparison）

- Lesson 9：Use the place value chart and metric measurement to compare decimals and answer comparison questions.
- Lesson 10：Use area models and the number line to compare decimal numbers, and record comparisons using <, >, and =.
- Lesson 11：Compare and order mixed numbers in various forms.
- 海报要点：从高位比起

D主题 十分位和百分位的加法（Topic D: Addition with Tenths and Hundredths）

- Lesson 12：Apply understanding of fraction equivalence to add tenths and hundredths.
- Lesson 13：Add decimal numbers by converting to fraction form.
- Lesson 14：Solve word problems involving the addition of measurements in decimal form.
- 海报要点：化成百分之几再加

E主题 用小数表示钱数（Topic E: Money Amounts as Decimal Numbers）

- Lesson 15：Express money amounts given in various forms as decimal numbers.
- Lesson 16：Solve word problems involving money.
- 海报要点：元、角、分
- 本地化：原书为美元、美分，本海报改为元、角、分。

第7模块 用乘法探索测量（Module 7: Exploring Measurement with Multiplication）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 单位换算表（Topic A: Measurement Conversion Tables）

- Lesson 1：Create conversion tables for length, weight, and capacity units using measurement tools, and use the tables to solve problems.
- Lesson 2：Create conversion tables for length, weight, and capacity units using measurement tools, and use the tables to solve problems.
- Lesson 3：Create conversion tables for units of time, and use the tables to solve problems.
- Lesson 4：Solve multiplicative comparison word problems using measurement conversion tables.
- Lesson 5：Share and critique peer strategies.
- 海报要点：时间和公制单位换算表
- 本地化：原书换算表含磅/盎司、英尺/英寸、加仑/夸脱/品脱，本海报只用公制单位和时间单位。

B主题 测量问题（Topic B: Problem Solving with Measurement）

- Lesson 6：Solve problems involving mixed units of capacity.
- Lesson 7：Solve problems involving mixed units of length.
- Lesson 8：Solve problems involving mixed units of weight.
- Lesson 9：Solve problems involving mixed units of time.
- Lesson 10：Solve multi-step measurement word problems.
- Lesson 11：Solve multi-step measurement word problems.
- 海报要点：混合单位；多步
- 本地化：原书含英制单位的测量问题，本海报只用公制。

C主题 带分数表示的测量（Topic C: Investigation of Measurements Expressed as Mixed Numbers）

- Lesson 12：Use measurement tools to convert mixed number measurements to smaller units.
- Lesson 13：Use measurement tools to convert mixed number measurements to smaller units.
- Lesson 14：Solve multi-step word problems involving converting mixed number measurements to a single unit.
- 海报要点：分数单位的测量
- 本地化：原书带分数测量以英制单位为主，本海报改为公制。

D主题 学年回顾（Topic D: Year in Review）

- Lesson 15：Create and determine the area of composite figures.
- Lesson 16：Create and determine the area of composite figures.
- Lesson 17：Practice and solidify Grade 4 fluency.
- Lesson 18：Practice and solidify Grade 4 vocabulary.
- 海报要点：游戏和复习

五年级（Grade 5）· 6 个模块 · 36 个 Topic

第1模块 位值与小数（Module 1: Place Value and Decimal Fractions）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 数位表上的乘法规律（Topic A: Multiplicative Patterns on the Place Value Chart）

- Lesson 1：Reason concretely and pictorially using place value understanding to relate adjacent base ten units from millions to thousandths.
- Lesson 2：Reason abstractly using place value understanding to relate adjacent base ten units from millions to thousandths.
- Lesson 3：Use exponents to name place value units, and explain patterns in the placement of the decimal point.
- Lesson 4：Use exponents to denote powers of 10 with application to metric conversions.
- 海报要点：乘除 10 的幂；指数

B主题 小数与位值（Topic B: Decimal Fractions and Place Value Patterns）

- Lesson 5：Name decimal fractions in expanded, unit, and word forms by applying place value reasoning.
- Lesson 6：Compare decimal fractions to the thousandths using like units, and express comparisons with >, <, =.
- 海报要点：千分位；展开式

C主题 小数的位值与近似（Topic C: Place Value and Rounding Decimal Fractions）

- Lesson 7：Round a given decimal to any place using place value understanding and the vertical number line.
- Lesson 8：Round a given decimal to any place using place value understanding and the vertical number line.
- 海报要点：近似到任意一位

D主题 小数加减（Topic D: Adding and Subtracting Decimals）

- Lesson 9：Add decimals using place value strategies, and relate those strategies to a written method.
- Lesson 10：Subtract decimals using place value strategies, and relate those strategies to a written method.
- 海报要点：相同数位相加减

E主题 小数乘法（Topic E: Multiplying Decimals）

- Lesson 11：Multiply a decimal fraction by single-digit whole numbers, relate to a written method through application of the area model and place value understanding, and explain the reasoning used.
- Lesson 12：Multiply a decimal fraction by single-digit whole numbers, including using estimation to confirm the placement of the decimal point.
- 海报要点：小数乘一位数

F主题 小数除法（Topic F: Dividing Decimals）

- Lesson 13：Divide decimals by single-digit whole numbers involving easily identifiable multiples using place value understanding and relate to a written method.
- Lesson 14：Divide decimals with a remainder using place value understanding and relate to a written method.
- Lesson 15：Divide decimals using place value understanding, including remainders in the smallest unit.
- Lesson 16：Solve word problems using decimal operations.
- 海报要点：小数除以一位数

第2模块 多位整数与小数运算（Module 2: Multi-Digit Whole Number and Decimal Fraction Operations）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 多位数乘法的口算（Topic A: Mental Strategies for Multi-Digit Whole Number Multiplication）

- Lesson 1：Multiply multi-digit whole numbers and multiples of 10 using place value patterns and the distributive and associative properties.
- Lesson 2：Estimate multi-digit products by rounding factors to a basic fact and using place value patterns.
- 海报要点：乘整十整百；估算

B主题 多位数乘法竖式（Topic B: The Standard Algorithm for Multi-Digit Whole Number Multiplication）

- Lesson 3：Write and interpret numerical expressions, and compare expressions using a visual model.
- Lesson 4：Convert numerical expressions into unit form as a mental strategy for multi-digit multiplication.
- Lesson 5：Connect visual models and the distributive property to partial products of the standard algorithm without renaming.
- Lesson 6：Connect area models and the distributive property to partial products of the standard algorithm with renaming.
- Lesson 7：Connect area models and the distributive property to partial products of the standard algorithm with renaming.
- Lesson 8：Fluently multiply multi-digit whole numbers using the standard algorithm and using estimation to check for reasonableness of the product.
- Lesson 9：Fluently multiply multi-digit whole numbers using the standard algorithm to solve multi-step word problems.
- 海报要点：标准算法

C主题 小数的多位数乘法（Topic C: Decimal Multi-Digit Multiplication）

- Lesson 10：Multiply decimal fractions with tenths by multi-digit whole numbers using place value understanding to record partial products.
- Lesson 11：Multiply decimal fractions by multi-digit whole numbers through conversion to a whole number problem and reasoning about the placement of the decimal.

- Lesson 12：Reason about the product of a whole number and a decimal with hundredths using place value understanding and estimation.
- 海报要点：小数乘两位数

D主题 乘法测量应用题（Topic D: Measurement Word Problems with Whole Number and Decimal Multiplication）

- Lesson 13：Use whole number multiplication to express equivalent measurements.
- Lesson 14：Use fraction and decimal multiplication to express equivalent measurements.
- Lesson 15：Solve two-step word problems involving measurement conversions.
- 海报要点：单位换算
- 本地化：原书部分应用题用英制单位，本海报只用公制。

E主题 多位数除法的口算（Topic E: Mental Strategies for Multi-Digit Whole Number Division）

- Lesson 16：Use divide by 10 patterns for multi-digit whole number division.
- Lesson 17：Use basic facts to approximate quotients with two-digit divisors.
- Lesson 18：Use basic facts to approximate quotients with two-digit divisors.
- 海报要点：除以整十；估商

F主题 部分商与整数除法（Topic F: Partial Quotients and Multi-Digit Whole Number Division）

- Lesson 19：Divide two- and three-digit dividends by multiples of 10 with single-digit quotients, and make connections to a written method.
- Lesson 20：Divide two- and three-digit dividends by two-digit divisors with single-digit quotients, and make connections to a written method.
- Lesson 21：Divide two- and three-digit dividends by two-digit divisors with single-digit quotients, and make connections to a written method.
- Lesson 22：Divide three- and four-digit dividends by two-digit divisors resulting in two- and three-digit quotients, reasoning about the decomposition of successive remainders in each place value.
- Lesson 23：Divide three- and four-digit dividends by two-digit divisors resulting in two- and three-digit quotients, reasoning about the decomposition of successive remainders in each place value.
- 海报要点：两位数除数

G主题 部分商与小数除法（Topic G: Partial Quotients and Multi-Digit Decimal Division）

- Lesson 24：Divide decimal dividends by multiples of 10, reasoning about the placement of the decimal point and making connections to a written method.
- Lesson 25：Use basic facts to approximate decimal quotients with two-digit divisors, reasoning about the placement of the decimal point.
- Lesson 26：Divide decimal dividends by two-digit divisors, estimating quotients, reasoning about the placement of the decimal point, and making connections to a written method.
- Lesson 27：Divide decimal dividends by two-digit divisors, estimating quotients, reasoning about the placement of the decimal point, and making connections to a written method.
- 海报要点：小数除以两位数

H主题 除法测量应用题（Topic H: Measurement Word Problems with Multi-Digit Division）

- Lesson 28：Solve division word problems involving multi-digit division with group size unknown and the number of groups unknown.

- Lesson 29：Solve division word problems involving multi-digit division with group size unknown and the number of groups unknown.
- 海报要点：多步问题
- 本地化：原书部分应用题用英制单位，本海报只用公制。

第3模块 分数加减法（Module 3: Addition and Subtraction of Fractions）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 相等的分数（Topic A: Equivalent Fractions）

- Lesson 1：Make equivalent fractions with the number line, the area model, and numbers.
- Lesson 2：Make equivalent fractions with sums of fractions with like denominators.
- 海报要点：1/2 = 2/4 = 3/6

B主题 画图找同单位（Topic B: Making Like Units Pictorially）

- Lesson 3：Add fractions with unlike units using the strategy of creating equivalent fractions.
- Lesson 4：Add fractions with sums between 1 and 2.
- Lesson 5：Subtract fractions with unlike units using the strategy of creating equivalent fractions.
- Lesson 6：Subtract fractions from numbers between 1 and 2.
- Lesson 7：Solve two-step word problems.
- 海报要点：长方形模型通分

C主题 计算找同单位（Topic C: Making Like Units Numerically）

- Lesson 8：Add fractions to and subtract fractions from whole numbers using equivalence and the number line as strategies.
- Lesson 9：Add fractions making like units numerically.
- Lesson 10：Add fractions with sums greater than 2.
- Lesson 11：Subtract fractions making like units numerically.
- Lesson 12：Subtract fractions greater than or equal to 1.
- 海报要点：通分相加减

D主题 拓展应用（Topic D: Further Applications）

- Lesson 13：Use fraction benchmark numbers to assess reasonableness of addition and subtraction equations.
- Lesson 14：Strategize to solve multi-term problems.
- Lesson 15：Solve multi-step word problems; assess reasonableness of solutions using benchmark numbers.
- Lesson 16：Explore part-to-whole relationships.
- 海报要点：多步分数问题

第4模块 分数和小数的乘除法（Module 4: Multiplication and Division of Fractions and Decimal Fractions）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 分数测量线图（Topic A: Line Plots of Fraction Measurements）

- Lesson 1：Measure and compare pencil lengths to the nearest , , and 2 4 8 of an inch, and analyze the data through line plots.

- 海报要点：测量数据中的分数
- 本地化：原书线图数据以英寸为单位，本海报改为厘米。

B主题 分数是除法（Topic B: Fractions as Division）

- Lesson 2：Interpret a fraction as division.
- Lesson 3：Interpret a fraction as division.
- Lesson 4：Use tape diagrams to model fractions as division.
- Lesson 5：Solve word problems involving the division of whole numbers with answers in the form of fractions or whole numbers.
- 海报要点：a ÷ b = a/b

C主题 整数乘分数（Topic C: Multiplication of a Whole Number by a Fraction）

- Lesson 6：Relate fractions as division to fraction of a set.
- Lesson 7：Multiply any whole number by a fraction using tape diagrams.
- Lesson 8：Relate a fraction of a set to the repeated addition interpretation of fraction multiplication.
- Lesson 9：Find a fraction of a measurement, and solve word problems.
- 海报要点：几分之几的整数

D主题 分数算式与应用题（Topic D: Fraction Expressions and Word Problems）

- Lesson 10：Compare and evaluate expressions with parentheses.
- Lesson 11：Solve and create fraction word problems involving addition, subtraction, and multiplication.
- Lesson 12：Solve and create fraction word problems involving addition, subtraction, and multiplication.
- 海报要点：写算式；条形图

E主题 分数乘分数（Topic E: Multiplication of a Fraction by a Fraction）

- Lesson 13：Multiply unit fractions by unit fractions.
- Lesson 14：Multiply unit fractions by non-unit fractions.
- Lesson 15：Multiply non-unit fractions by non-unit fractions.
- Lesson 16：Solve word problems using tape diagrams and fraction-by-fraction multiplication.
- Lesson 17：Relate decimal and fraction multiplication.
- Lesson 18：Relate decimal and fraction multiplication.
- Lesson 19：Convert measures involving whole numbers, and solve multi-step word problems.
- Lesson 20：Convert mixed unit measurements, and solve multi-step word problems.
- 海报要点：面积模型

F主题 乘法是缩放（Topic F: Multiplication with Fractions and Decimals as Scaling and Word Problems）

- Lesson 21：Explain the size of the product, and relate fraction and decimal equivalence to multiplying a fraction by 1.
- Lesson 22：Compare the size of the product to the size of the factors.
- Lesson 23：Compare the size of the product to the size of the factors.
- Lesson 24：Solve word problems using fraction and decimal multiplication.
- 海报要点：积比乘数大还是小

G主题 分数和小数的除法（Topic G: Division of Fractions and Decimal Fractions）

- Lesson 25：Divide a whole number by a unit fraction.
- Lesson 26：Divide a unit fraction by a whole number.
- Lesson 27：Solve problems involving fraction division.
- Lesson 28：Write equations and word problems corresponding to tape and number line diagrams.
- Lesson 29：Connect division by a unit fraction to division by 1 tenth and 1 hundredth.
- Lesson 30：Divide decimal dividends by non-unit decimal divisors.
- Lesson 31：Divide decimal dividends by non-unit decimal divisors.
- 海报要点：单位分数与整数相除

H主题 解读算式（Topic H: Interpretation of Numerical Expressions）

- Lesson 32：Interpret and evaluate numerical expressions including the language of scaling and fraction division.
- Lesson 33：Create story contexts for numerical expressions and tape diagrams, and solve word problems.
- 海报要点：读懂并写出算式

第5模块 体积和面积中的加法与乘法（Module 5: Addition and Multiplication with Volume and Area）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 体积的概念（Topic A: Concepts of Volume）

- Lesson 1：Explore volume by building with and counting unit cubes.
- Lesson 2：Find the volume of a right rectangular prism by packing with cubic units and counting.
- Lesson 3：Compose and decompose right rectangular prisms using layers.
- 海报要点：单位立方体；层

B主题 体积与运算（Topic B: Volume and the Operations of Multiplication and Addition）

- Lesson 4：Use multiplication to calculate volume.
- Lesson 5：Use multiplication to connect volume as packing with volume as filling.
- Lesson 6：Find the total volume of solid figures composed of two non-overlapping rectangular prisms.
- Lesson 7：Solve word problems involving the volume of rectangular prisms with whole number edge lengths.
- Lesson 8：Apply concepts and formulas of volume to design a sculpture using rectangular prisms within given parameters.
- Lesson 9：Apply concepts and formulas of volume to design a sculpture using rectangular prisms within given parameters.
- 海报要点：长 × 宽 × 高；组合体

C主题 分数边长的长方形面积（Topic C: Area of Rectangular Figures with Fractional Side Lengths）

- Lesson 10：Find the area of rectangles with whole-by-mixed and whole-by-fractional number side lengths by tiling, record by drawing, and relate to fraction multiplication.
- Lesson 11：Find the area of rectangles with mixed-by-mixed and fraction- by-fraction side lengths by tiling, record by drawing, and relate to fraction multiplication.

- Lesson 12：Measure to find the area of rectangles with fractional side lengths.
- Lesson 13：Multiply mixed number factors, and relate to the distributive property and the area model.
- Lesson 14：Solve real-world problems involving area of figures with fractional side lengths using visual models and/or equations.
- Lesson 15：Solve real-world problems involving area of figures with fractional side lengths using visual models and/or equations.
- 海报要点：分数乘分数的面积

D主题 平面图形的画、分析和分类（Topic D: Drawing, Analysis, and Classification of Two-Dimensional Shapes）

- Lesson 16：Draw trapezoids to clarify their attributes, and define trapezoids based on those attributes.
- Lesson 17：Draw parallelograms to clarify their attributes, and define parallelograms based on those attributes.
- Lesson 18：Draw rectangles and rhombuses to clarify their attributes, and define rectangles and rhombuses based on those attributes.
- Lesson 19：Draw kites and squares to clarify their attributes, and define kites and squares based on those attributes.
- Lesson 20：Classify two-dimensional figures in a hierarchy based on properties.
- Lesson 21：Draw and identify varied two-dimensional figures from given attributes.
- 海报要点：四边形层级

第6模块 坐标平面中的问题解决（Module 6: Problem Solving with the Coordinate Plane）

- 来源：Teacher Edition PDF · EngageNY 模块页（档案）

A主题 坐标系（Topic A: Coordinate Systems）

- Lesson 1：Construct a coordinate system on a line.
- Lesson 2：Construct a coordinate system on a plane.
- Lesson 3：Name points using coordinate pairs, and use the coordinate pairs to plot points.
- Lesson 4：Name points using coordinate pairs, and use the coordinate pairs to plot points.
- Lesson 5：Investigate patterns in vertical and horizontal lines, and interpret points on the plane as distances from the axes.
- Lesson 6：Investigate patterns in vertical and horizontal lines, and interpret points on the plane as distances from the axes.
- 海报要点：数线与坐标轴；有序数对

B主题 坐标平面中的规律（Topic B: Patterns in the Coordinate Plane and Graphing Number Patterns from Rules）

- Lesson 7：Plot points, use them to draw lines in the plane, and describe patterns within the coordinate pairs.
- Lesson 8：Generate a number pattern from a given rule, and plot the points.
- Lesson 9：Generate two number patterns from given rules, plot the points, and analyze the patterns.
- Lesson 10：Compare the lines and patterns generated by addition rules and multiplication rules.
- Lesson 11：Analyze number patterns created from mixed operations.

- Lesson 12：Create a rule to generate a number pattern, and plot the points.
- 海报要点：数列规则与图像

C主题 在坐标平面上画图形（Topic C: Drawing Figures in the Coordinate Plane）

- Lesson 13：Construct parallel line segments on a rectangular grid.
- Lesson 14：Construct parallel line segments, and analyze relationships of the coordinate pairs.
- Lesson 15：Construct perpendicular line segments on a rectangular grid.
- Lesson 16：Construct perpendicular line segments, and analyze relationships of the coordinate pairs.
- Lesson 17：Draw symmetric figures using distance and angle measure from the line of symmetry.
- 海报要点：平行线、垂直线、对称

D主题 坐标平面中的问题（Topic D: Problem Solving in the Coordinate Plane）

- Lesson 18：Draw symmetric figures on the coordinate plane.
- Lesson 19：Plot data on line graphs and analyze trends.
- Lesson 20：Use coordinate systems to solve real-world problems.
- 海报要点：线和点的关系

E主题 多步应用题（Topic E: Multi-Step Word Problems）

- Lesson 21：Make sense of complex, multi-step problems, and persevere in solving them. Share and critique peer solutions.
- Lesson 22：Make sense of complex, multi-step problems, and persevere in solving them. Share and critique peer solutions.
- Lesson 23：Make sense of complex, multi-step problems, and persevere in solving them. Share and critique peer solutions.
- Lesson 24：Make sense of complex, multi-step problems, and persevere in solving them. Share and critique peer solutions.
- Lesson 25：Make sense of complex, multi-step problems, and persevere in solving them. Share and critique peer solutions.
- 海报要点：综合应用

F主题 小学数学回顾（Topic F: The Years In Review: A Reflection on A Story of Units）

- Lesson 26：Solidify writing and interpreting numerical expressions.
- Lesson 27：Solidify writing and interpreting numerical expressions.
- Lesson 28：Solidify fluency with Grade 5 skills.
- Lesson 29：Solidify the vocabulary of geometry.
- Lesson 30：Solidify the vocabulary of geometry.
- Lesson 31：Explore the Fibonacci sequence.
- Lesson 32：Explore patterns in saving money.
- Lesson 33：Design and construct boxes to house materials for summer use.
- Lesson 34：Design and construct boxes to house materials for summer use.
- 海报要点：回顾整个小学数学